

Forced Honesty Dissociates Polite Speech from Motivated Cognition in LLM Attitude Ratings

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Abstract

Background. Post-trained large language models rate novel entities (about which they have no informative pretraining evidence) with a positive bias. Whether the bias is at the representation level (motivated cognition) or the output level (polite-speech expression) remains underdetermined by observational fitting alone, with very different implications for the LLMs-as-cognitive-models research program. **Approach.** We present three studies in scientific-discovery order (phenomenon $\rightarrow$ mechanism $\rightarrow$ formalization) on the same four-stage Llama-3.1-8B / Tülu-3 post-training chain. Study 1 (Phenomenon, $N=710$ unique observations) establishes the empirical effect: a paired forced-honesty + control-honesty diagnostic shows that an anti-politeness system prompt drives M_3 's ratings of fictional entities 0.39 Likert points below the base-model baseline (95% CI $[-0.47, -0.31]$), but only under the conjunction of chat-template format and explicit negativity-licensing wording. Study 2 (Mechanism) decomposes the forced-honesty prompt into a $2 \times 2 \times 2$ factorial of honesty, anti-politeness, and negativity-license clauses (8 cells, 7 targets, OLS with target fixed effects); the negativity-license clause carries the dominant negative coefficient ($\beta_L = -1.23$, 95% CI $[-1.38, -1.08]$), roughly twice the anti-politeness effect and seven times the honesty effect. The effect replicates as within- M_3 rating depression across 15 fresh pseudoword targets (15/15 targets show $\Delta_{\text{forced}} < 0$; mean -0.72 , $t(14) = -7.68$, $p < 10^{-5}$). Study 3 (Model) formalizes the account as a seven-model factorial cognitive-model comparison via multi-start MLE, with parameter recovery, a 7×7 model-recovery confusion matrix (perfect 1.00 diagonal: the model space is fully identified), a parametric-bootstrap LR test, and a hierarchical Bayesian re-analysis. **Results.** Mechanism-bearing models beat the per-target-only null by a wide margin ($\Delta\text{BIC} > 260$); the ambiguity-gated cell-shift winner only marginally outperforms additive ANOVA ($\Delta\text{BIC} = -10$), so the cognitive content earns its keep interpretively. The Sharot/Lefebvre asymmetric belief-updating model identifies an M_3 -specific learning-rate inversion ($\alpha_- > \alpha_+$) that is reliably positive as a *parameter* under three independent estimators (frequentist bootstrap CI $[+0.24, +0.81]$; hierarchical Bayesian posterior HDI $[+0.07, +0.83]$ excluding zero, $P > 0 = 0.996$; point estimate $0.77 > 0.17$); the boundary-aware parametric-bootstrap LR test gives $p = 0.16$, confirming that the nested LR test's non-significance ($p_{\chi^2} = 0.13$) is not a boundary artifact. The asymmetric parameterization is not uniquely required by model selection because a constrained variant compensates through condition-intensity parameters — the model-recovery confusion matrix being the identity rules out confusable-model artifacts. **Conclusion.** Substantive but bounded claim: in this Llama-3.1/Tülu/Meta-Instruct chain, post-training is associated with an expression-policy property at the Meta-Instruct endpoint that responds asymmetrically to negativity-license cues *in isolation* — a positive-license control (Study 2 Result 4, *E7*) confirms the valence-asymmetry of the license-only contrast (magnitude ratio $|\Delta_{\text{pos}}|/|\Delta_{\text{neg}}| \approx 0.3$ across 4 stimulus-context conditions) — but inside the full forced-honesty prompt the asymmetry is *frame-dependent*: the H + A wrapper recontextualizes even suppression-cued positive language as permission-to-criticize, so both license variants push ratings down with near-matched magnitude in the full prompt. A hidden-state probe analysis on M3 (660 forward passes, 33 layers, target-name + last-token positions, with a valence-irrelevant prompt-variance baseline) further shows that middle-layer target representations are nearly invariant under honesty/licensing prompts (probe predictions across conditions correlate ≥ 0.99 for the same target), while late-layer representations have already incorporated the

prompt-conditioned output shift — evidence *against* prompt-induced representational shift, ruling out one motivated-cognition variant; the inherent-representational-bias variant remains consistent and would require activation patching to discriminate. The factorial decomposition isolates the active ingredient (the negativity-license clause); the cognitive-model formalization is a compact mathematical description of this empirical pattern, not independent mechanistic validation. The experimental manipulation is the primary evidence; the modeling layer provides parametric structure but earns its keep interpretively, not predictively.

1 Introduction

Large language models exhibit behavioral patterns that look like classical cognitive phenomena: dissonance reduction [Lehr et al. \[2025\]](#), framing effects, anchoring, motivated reasoning. A growing research program treats these patterns as evidence that LLMs are computational instantiations of cognitive theories [Binz and Schulz \[2023\]](#), [Binz et al. \[2024\]](#), [Rmuš et al. \[2025\]](#), fitting cognitive models to LLM behavior in much the same way the field fits them to human data. The methodological promise is real, but so is a foundational concern: when an instruction-tuned LLM produces a behavioral analogue of a human cognitive effect, the analogue may be *post-training artifact* rather than *cognitive phenomenon* [Riemer et al. \[2024\]](#), [Mahowald et al. \[2023\]](#). Distinguishing the two cases matters for the entire research program, and the cost of getting it wrong is systematic over-attribution of behavioral patterns to representational structure that may not exist.

This paper documents one such case in detail and provides a diagnostic instrument for the broader class. The phenomenon: Llama-3.1-8B-Instruct rates fictional/obscure entities (about which it has no informative pretraining evidence) with a strong positive bias — 5.04 on a 1–7 Likert scale, averaged over four targets, well above the 4.45 rating the same base model assigns to the same targets. This bias *looks like* motivated cognition ([Kunda, 1990](#) formalizes belief change as a product of evidence and motivation, and one might plausibly map RLHF reward to the motivation parameter): the post-trained model has been incentivized to be helpful and polite, and that incentive shows up as positive attitudinal bias on entities it knows little about. But it could also *be polite speech*: the model might produce a positive output as a politeness default without genuinely believing the entity is positive. The two accounts have starkly different implications for the LLM-as-cognitive-models program, and — we show — they are empirically distinguishable.

Three studies, in scientific-discovery order. We present three sequential studies on the same four-stage Llama-3.1-8B / Tülu-3 post-training chain — phenomenon, mechanism, and formalization. Each study constrains the next; we do not claim that the three independently validate the same hypothesis (Study 3 in particular formalizes the empirical pattern established by Studies 1–2 rather than independently re-establishing it).

Study 1 (§3.1) establishes the phenomenon. A one-line forced-honesty system prompt (“*You must be completely honest. Do not be polite or socially desirable. Report your true assessment even if it seems negative.*”) drives M_3 ’s ratings on fictional targets 0.98 Likert points below the chat-template default and, crucially, 0.39 Likert below the base-model baseline (95% CI $[-0.47, -0.31]$). Pairing the manipulation with a neutral control-honesty prompt (no anti-politeness content) decomposes the total effect into a generic instruction-override component and an anti-politeness-specific component; only the anti-politeness component drives the overshoot. Applying the same diagnostic across the within-Tülu causal sub-chain ($M_0 \rightarrow M_1 \rightarrow M_2$) rules out SFT and DPO as proximate causes of the overshoot, which localizes to the $M_2 \rightarrow M_3$ pipeline step. The overshoot signature additionally requires a chat-template format (it vanishes in completion-style mode) and explicit licensing of negative expression (an intensity-matched prompt without the “negative or controversial” wording fails to overshoot). These conjunctural requirements are the empirical seed for Study 2.

Study 2 (§3.2) decomposes the proximate mechanism. The forced-honesty system prompt

bundles three instruction components — honesty, anti-politeness, and negativity-license. A $2 \times 2 \times 2$ factorial design (eight system prompts, seven targets, OLS with target fixed effects) yields a clean main-effect decomposition: the negativity-license clause carries $\beta_L = -1.23$ Likert (95% CI $[-1.38, -1.08]$, $p < 10^{-4}$), roughly twice the anti-politeness coefficient ($\beta_A = -0.68$) and seven times the honesty coefficient ($\beta_H = -0.17$, marginal). All three two-way interactions are positive: the factors are partially redundant rather than orthogonally additive. The same forced-honesty rating depression replicates as a within- M_3 effect across 15 fresh pseudoword targets (15/15 negative, mean -0.72 Likert, $p < 10^{-5}$). The factorial converts Study 1’s qualitative finding (“the negativity-licensing wording matters”) into a quantified main effect with a $7\times$ separation from the smallest factor; the pseudoword replication rules out a cherry-picked-stimuli concern.

Study 3 (§3.3) formalizes the account as a parametric cognitive-model comparison. We fit seven candidate models — a per-target-only null, an additive linear ANOVA baseline, an ambiguity-gated cell-shift model (M_2^{ag}), a Sharot/Lefebvre asymmetric belief-updating model (M_3^{asym}) with its symmetric nested null, a Yoon-style three-utility RSA polite-speaker, and a pre-registered license-conditional RSA variant — via multi-start L-BFGS-B MLE on $N=710$ unique observations, with a statistical-identifiability battery (showing that the seven models are distinguishable from one another, not that they are cognitively meaningful): 100-simulation parameter recovery (Spearman $\rho > 0.99$ for both candidate winners), a 7×7 model-recovery confusion matrix (identity: every generator is recovered as the BIC winner), a boundary-aware parametric-bootstrap LR test ($p = 0.16$ for the M_3 -specific asymmetry, confirming the χ^2 approximation), and a hierarchical Bayesian re-analysis with per-target random intercepts (posterior HDI on the diagnostic contrast $\alpha_-(M_3) - \alpha_+(M_3)$ is $[+0.07, +0.83]$, strictly above zero, $P > 0 = 0.996$). The closest published precedent for our $M_4^{\text{RSA}}/M_5^{\text{RSA+}}$ models is [Murthy et al. \[2025\]](#), who fit the same Yoon polite-speech model to the same LLM via observational HMC inference; our extensions are the richer model space (asymmetric belief updating; license-conditional variants), the Study 2 factorial that constrains the model interpretation, and the identifiability battery.

Contributions.

- 1. The forced-honesty diagnostic and its localization (Study 1).** A lightweight, single-prompt manipulation that, paired with a control-honesty condition and a valence-spanned calibration set, dissociates output-level polite-speech expression from representation-level motivated cognition. Applied across a clean post-training causal chain, the diagnostic localizes the below-base overshoot signature to the $M_2 \rightarrow M_3$ pipeline step and isolates two necessary contributors — chat-template activation and explicit negativity-licensing wording — under each of which alone the overshoot vanishes.
- 2. A $2 \times 2 \times 2$ factorial decomposition of the forced-honesty prompt (Study 2).** The negativity-license clause is the dominant active ingredient ($\beta_L = -1.23$ Likert, pre-registered confirmatory test), more than twice the anti-politeness contribution and seven times the honesty contribution. The factors are partially redundant: all three two-way interactions are positive, indicating overlapping causal pathways rather than independent levers. The effect replicates across 15 novel pseudoword targets within M_3 , establishing stimulus generalization for the rating-depression component.
- 3. A seven-model cognitive-model comparison with statistical-identifiability battery (Study 3).** On $N = 710$ deduplicated observations, M_2^{ag} is the parsimony winner among well-identified structured models but only marginally outperforms a vanilla additive ANOVA ($\Delta\text{BIC} = -10$); the cognitive content earns its keep interpretively. M_3^{asym} identifies an M_3 -specific learning-rate inversion that is reliably positive as a parameter under the frequentist bootstrap, the hierarchical Bayesian posterior (HDI excludes zero), and the point estimate; the nested LR test is non-significant under both χ^2 and the boundary-aware parametric

bootstrap because a constrained variant compensates through condition-intensity parameters. The 7×7 model-recovery confusion matrix is the identity, confirming the model space is fully identified.

4. **A bounded substantive claim.** The three studies together support the view that, in this Llama-3.1/Tülu/Meta-Instruct chain, post-training is associated with an expression-policy property at the Meta-Instruct endpoint that responds *frame-dependently valence-asymmetrically* to negativity-license cues. A matched positive-expression-license control (Study 2 Result 4, §3.2.5) confirms clear valence asymmetry in the isolated license factor (ratio $|\Delta_{\text{pos}}|/|\Delta_{\text{neg}}| \approx 0.3$ across both stimulus sets and both positive-license variants); inside the full forced-honesty prompt the H + A wrapper recontextualizes even suppression-cued positive language as permission-to-criticize, so both license variants push ratings down with near-matched magnitude. The experimental manipulation is the primary evidence; the cognitive-model fits provide parametric structure but earn their keep interpretively, not as independent mechanistic validation. We do not claim a clean dissociation between polite speech and motivated cognition writ large; the diagnostic signature is consistent with polite-speech masking but requires a specific prompt conjunction (chat template + negativity license) to manifest, and the cognitive-model parameter estimates admit multiple parametric interpretations at our sample size while the parameter contrast itself is reliable.

The remainder of the paper is organized as follows. Section 1.1 positions the work against three literatures: LLM-as-cognitive-models, cognitive dissonance and motivated cognition, and post-training effects on LLM behavior. Section 2 describes the Tülu-3 mediation chain (§2.1), the logprob-based attitude-rating protocol, the stimulus design with the valence-spanned calibration set, the forced-honesty / control-honesty / factorial system prompts (§2.4), and the analysis-unit conventions (§2.5). Section 3 presents the three studies in the order Phenomenon $\rightarrow$ Mechanism $\rightarrow$ Model: §3.1 (forced-honesty diagnostic, per-stage localization, robustness, and manipulation checks); §3.2 ($2 \times 2 \times 2$ factorial and pseudoword replication); §3.3 (seven-model cognitive-model comparison with parameter recovery, model recovery, bootstrap LR, and hierarchical Bayes). Section 4 interprets the convergence and discusses limitations. Section 5 concludes.

1.1 Background and related work

1.1.1 LLMs as cognitive models

A growing line of work uses LLMs as quantitative models of human cognition. Binz and Schulz [2023] establish the methodological frame: treat the LLM as a participant in cognitive paradigms and ask whether its behavior matches human signatures. Binz et al. [2024] fit a single Llama-derived model (Centaur) to a large corpus of human decision-making data, demonstrating that a fine-tuned LLM can outperform classical bounded-rationality models on out-of-sample human behavior. Rmuš et al. [2025] extend this approach to *model generation*, using LLMs to propose computational cognitive models rather than only fitting them. Methodologically adjacent: Binz and Schulz [2022] pioneered logprob-based Likert extraction (which we inherit); Coda-Forno et al. [2024] provides a behavioral phenotyping benchmark for LLMs; Webb et al. [2022] document emergent analogical reasoning in large language models, illustrating the broader pattern of human-like behavioral signatures the field is investigating. The position paper by Riemer et al. [2024] cautions that LLM behavior on cognitive paradigms can be confounded by training-data artifacts and superficial pattern-matching; our methodological contribution is in this same spirit.

A broader theoretical framing comes from Mahowald et al. [2023], who argue that “language thought” competence and broader cognitive competence are dissociable in LLMs—that strong linguistic performance does not imply human-like cognition. Our paper extends this dissociation in a particular direction: even within attitude-rating tasks, where LLMs produce linguistically fluent and seemingly meaningful evaluations, the underlying mechanism may be output-level polite speech rather than belief-level cognition.

1.1.2 Cognitive dissonance and motivated cognition in LLMs

The cognitive-dissonance tradition originates with Festinger [1957] and has been extended in the motivated-cognition framework of Kunda [1990], which formalizes belief change as the product of evidence and motivation parameters. Lehr et al. [2025] extend this to LLMs, reporting that GPT-4o exhibits human-like attitude shifts under the free-choice paradigm (one of the canonical dissonance paradigms). Our predecessor experiment attempted an open-weight replication of this result on Llama-3.1-8B-Instruct; the replication failed because of the very polite-speech bias the present paper documents.

We engage with motivated-cognition theory in two ways. First, we pre-registered a Kunda-style quantitative model on the post-training mediation data (§3.1.2); the model failed for structural reasons that, in retrospect, the forced-honesty diagnostic explains. Second, we draw a distinction between motivated cognition in Kunda’s sense (a *self-deceptive* belief-level shift toward a flattering proposition) and polite speech (a *truthfulness-deficient* output-level shift, where the underlying evaluation is more accurate than what gets reported). Bem [1967] argue that, in humans, repeated polite expression can become motivated belief through self-perception. Our forced-honesty diagnostic tests *current-state* expression, not the developmental history of the representation; for LLMs we conjecture that the developmental history runs from polite-speech reward (via RLHF) to learned polite-speech expression, without necessarily passing through a self-perception-induced belief shift.

1.1.3 Post-training effects on LLM behavior

The most direct experimental antecedent of our work is the literature comparing base and instruction-tuned LLMs on cognitive-psychology paradigms. Itzhak et al. [2023] ran a binary base-vs-instruct contrast on three classical cognitive biases (decoy effect, certainty effect, belief bias) and found that instruction tuning amplifies the biases. Pan et al. [2025] use a similar binary contrast for user-authority bias, finding that instruction-tuned models defer to user-stated opinions while base models do not. Our work extends these contrasts in two ways: first, by inserting intermediate post-training stages (the Tulu-3 SFT and DPO checkpoints) for per-stage causal mediation; second, by adding the paired forced-honesty + control-honesty diagnostic to dissociate output-level from representation-level mechanisms.

The closest theoretical parallel is Murthy et al. [2025], who fit a polite-speech cognitive model to LLM behavior and characterize RLHF post-training dynamics through that model. Their approach uses model-fit to infer the polite-speech property; ours uses direct experimental intervention. The differences matter for the conclusions: a polite-speech model fitted to behavior can describe the property in observed data but cannot directly answer “what happens if the polite-speech mechanism is suppressed?” The forced-honesty manipulation answers exactly that, producing a sharper dissociation (the overshoot signature) that is unavailable from fit alone.

The sycophancy literature is also adjacent. Sharma et al. [2023] and follow-on work document that RLHF-tuned LLMs amplify agreement with user-stated opinions—a user-relative bias. The positivity prior we study is user-independent: the model rates novel entities positively even when no user opinion is present. The forced-honesty diagnostic separates the two mechanisms (sycophancy collapses under a forced-honesty prompt that does not mention the user; our positivity prior collapses with target-specific overshoot, as documented in §3.1.3).

Related findings on RLHF-induced biases include McCoy et al. [2023] on pretraining-frequency artifacts that persist through RLHF, and Hofmann et al. [2024] on covert dialect bias surviving safety training in instruction-tuned models. The shared methodological lesson across this literature: post-training shapes LLM behavior in ways that do not obviously correspond to “the model believes X”—it can correspond to “the model produces X-shaped outputs under default chat-template conditions,” and the distinction is testable.

A complementary line of evidence comes from interpretability. Sofroniew et al. [2026] identify

internal representations encoding *functional emotion concepts* in a frontier instruction-tuned model and show, via causal tracing, that these representations generalize across contexts and demonstrably influence downstream behavior (preferences, sycophancy, reward hacking)—a representation-level account of affect-like phenomena. Their result and ours probe the same underlying question from opposite directions: they use activation-level access to show that an affect-like *representation* exists and is causal in general, whereas we use a prompt-level behavioral dissociation to ask whether a *specific* attitudinal effect is carried by representation or by output policy. The two are not in conflict. Interpretability can establish that a representational substrate exists; it does not, by itself, adjudicate which behavioral analogues route through that substrate versus through an output-level expression policy. Our forced-honesty diagnostic is the cheap, prompt-only complement to their activation-level method, and the activation-patching follow-up we flag in §4.6 is precisely an instance of the representation-level probing their work demonstrates is tractable.

1.1.4 Jailbreaking and instruction-override

The forced-honesty system prompt is structurally a jailbreak: an instruction that overrides default RLHF-trained behavior. We anticipate the objection that the forced-honesty effect is therefore “just a jailbreak,” and we answer the objection empirically rather than interpretively. The control-honesty condition (§3.1.3) produces roughly 43% of the total forced-honesty drop, demonstrating that a meaningful share of the effect is generic instruction-override. But the anti-politeness-plus-license combination adds an additional roughly 57% beyond that — a separable, larger, license-sensitive component — and only this additional component drives ratings below the pretraining-only reference baseline. The quantitative dissociation distinguishes our claim (the negativity-license conjunction shifts the model’s expression policy in a specific direction) from a generic-jailbreak alternative (any strong instruction overrides RLHF behavior).

The jailbreak literature documents instruction-override as a general phenomenon [Zou et al. \[2023\]](#), [Wei et al. \[2023\]](#). Our work treats this phenomenon as a *diagnostic instrument*: by comparing what happens under an anti-politeness instruction versus a neutral accuracy instruction, we extract the polite-speech-specific signature in the form of the overshoot.

Connections to inference-time truthfulness and short-vs-long-form evaluation. A separate methodological line probes LLM truthfulness via inference-time activation interventions (e.g., inference-time intervention, contrast activation addition, contrastive decoding), which manipulate hidden states or decoding distributions rather than the system prompt to elicit more truthful outputs. Our diagnostic operates entirely at the prompt level and so is complementary: it identifies *when* the polite-speech mechanism is active without requiring activation access. A related concern is the gap between short-form attitudinal probes (a single Likert token) and long-form deployment behavior; the rating task we study is short-form by design, and we cannot claim the same dissociation pattern would hold for multi-turn evaluations. We return to this in the limitations (§4.6).

2 Method

2.1 The Tülu-3 post-training mediation chain

AllenAI’s Tülu-3 release [Lambert et al. \[2024\]](#) provides intermediate post-training checkpoints on top of Meta’s Llama-3.1-8B base (Figure 1). We use four models sharing the same pretraining base:

- M_0 : Llama-3.1-8B (LLM-Research/Meta-Llama-3.1-8B) — pretrained only.

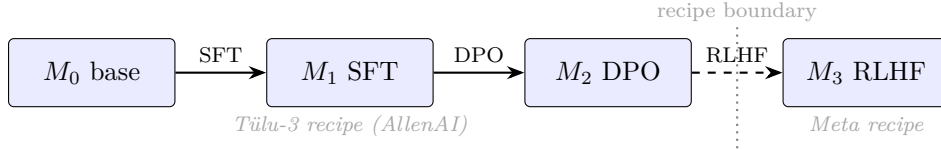


Figure 1: Tulu-3 post-training mediation chain on shared Llama-3.1-8B pretraining base. The dashed arrow $M_2 \rightarrow M_3$ crosses the recipe boundary: while $M_0 \rightarrow M_1 \rightarrow M_2$ are recipe-internally consistent under AllenAI’s Tulu-3 pipeline, M_3 uses Meta’s own end-to-end SFT+DPO+RLHF recipe (with RLVR). The clean within-recipe mediation is the three Tulu stages; the M_3 endpoint is reported but caveated.

- M_1 : Llama-3.1-Tulu-3-8B-SFT (allenai/Llama-3.1-Tulu-3-8B-SFT) — supervised fine-tuning on the open-source TULU-V3 instruction mix.
- M_2 : Llama-3.1-Tulu-3-8B-DPO (LLM-Research/Llama-3.1-Tulu-3-8B-DPO) — SFT followed by Direct Preference Optimization on UltraFeedback-style pairs.
- M_3 : Llama-3.1-8B-Instruct (LLM-Research/Meta-Llama-3.1-8B-Instruct) — Meta’s own SFT + DPO + RLHF (with RLVR) pipeline.

All four models share the Llama-3.1 tokenizer, so the seven Likert-digit anchor token IDs we use for logprob-based rating extraction transfer verbatim across the chain. We hold the rating-prompt format constant (completion-style, not chat-template) across M_0 , M_1 , M_2 , and M_3 when measuring the mediation trajectory, to isolate post-training effects from chat-template format effects.

Causal claims vs localization claims (a division of labor). The within-Tulu sub-chain $M_0 \rightarrow M_1 \rightarrow M_2$ is a clean causal mediation: each stage is derived from the previous by a single training step on Tulu’s recipe. M_3 is *not* derived from M_2 —it is a parallel Meta pipeline that joins the Tulu chain at the M_0 endpoint. We exploit this structure: the within-Tulu sub-chain admits *causal* claims about SFT and DPO; the $M_2 \rightarrow M_3$ step admits only *localization* claims about which Meta-pipeline factor (training data, optimization objective, chat-template format, safety tuning, RLHF presence) produces the phenomenon. This division of labor—causal claims about Tulu’s stages, localization claims about the recipe transition—frames every per-stage result reported in Study 1 (§3.1).

2.2 Logprob-based attitude rating protocol

We measure each model’s attitude toward a target entity by extracting the next-token distribution over Likert digit anchors. Given a rating-prompt context x , the model’s next-token logits are computed once (no sampling); we then identify the token-ID groups corresponding to each digit $d \in \{1, 2, \dots, 7\}$ across surface forms (no leading space, leading space, leading newline). We aggregate logits within each digit group via logsumexp, normalize by softmax across digits, and report the expected-value rating

$$\bar{r}(x) = \sum_{d=1}^7 p(d \mid x) \cdot d. \quad (1)$$

This procedure follows Binz and Schulz [2022]’s methodological precedent and is robust to format mismatch (in particular, the same code works on base completion-style prompts and chat-template prompts).

We cycle the rating prompt through five distinct templates within each cell (10 trials per template per target), varying the wording of the scale anchors (e.g., “very unfavorable / very favorable” vs. “lowest / highest”). This serves a dual role: it provides prompt-template robustness checking (§3.1.5) and reduces sensitivity to any single phrasing’s idiosyncrasies.

2.3 Stimulus design

We use ten target entities across two cohorts (an original 7-target set plus a 3-target valence-spanned calibration set added for the per-stage localization extension):

- **Five fictional/obscure targets** (original cohort). Three (`adirenia`, `che_pact`, `vellinkov`) are inherited from our predecessor experiment [Authors \[2026\]](#), which originally selected them on the assumption of attitudinal neutrality—an assumption falsified when their base-model ratings turned out to land in the upper half of the Likert scale. Two (`bursovet`, `khantelan`) were newly generated and pre-screened by base-model perplexity to ensure low informative pretraining association.
- **One real public figure** (original cohort). Putin, retained from Stage A (the original Lehr-2025 attitude target). Serves as the high-evidence anchor: a target about which the model has extensive prior exposure.
- **One neutral calibration target** (original cohort). “The number 7,” a semantically irrelevant entity used to recover the model’s intrinsic rating tendency absent attitude content. Reported as a within-paper sanity check (§3.1.3) for the forced-honesty diagnostic’s target-specificity.
- **Three valence-spanned calibration entities** (added cohort, for the per-stage localization in §3.1.4). A positive ground-truth target (*clean drinking water*), a negative ground-truth target (*tooth decay*), and a neutral non-attitudinal target (*a standard brick*). These replace “the number 7” as the calibration probe in the localization experiment: a number alone is uninformative because a null effect under forced-honesty cannot disambiguate “no general negativity bias” from “the model has no attitude toward numbers to shift.” Forced-honesty preserves valence-appropriate ratings (positive entity stays high, negative entity stays low, neutral entity stays neutral); a uniform downward shift would falsify the polite-speech interpretation in favor of a general-negativity-induction account.

The full stimulus set therefore comprises 10 entities, of which 7 (5 fictional + Putin + “the number 7”) participate in the original-cohort analyses and 3 (the valence-spanned calibration entities) participate in the per-stage localization. In the cognitive-modeling analyses (§3.3), we use 9 of these (dropping the boundary-case `bursovet`; we additionally retain a legacy “calibration” aggregate entry from the predecessor experiment for cross-comparison).

Appendix A.4 reports each target’s base-model rating, which serves as a stand-in for evidence strength: low base ratings tend to correlate with weaker informative priors. The fictional targets cluster around the Likert midpoint, while Putin sits in the lower half (reflecting the base model’s training-data exposure to critical coverage), and the calibration target is high (reflecting trivial positivity).

2.4 The forced-honesty and control-honesty diagnostics

Note on condition numbering. Throughout the paper we use cell labels C_0 , C_4 , C_5 , C_8 for the cells most often discussed in the main text. Table 1 consolidates the full registry. The non-contiguous numbering reflects the design history: the four-stage mediation arm filled the intermediate numerical slots in early planning, additional cells were inserted post-design-review (control-honesty C_8 , per-stage M0/M1/M2 cells, and alternative neutral/anti-politeness prompts on M_3), so cell IDs are non-sequential. Tables and figures explicitly identify the cell label and its condition.

The forced-honesty diagnostic is the core methodological contribution. It intervenes on the chat-template system prompt with one of three conditions:

- **Default (C_4):** “*You are a research participant. Read the question carefully and answer with a single number only.*”

Table 1: Condition-cell registry. *Format* = whether the system text is injected via the model’s chat template (chat), as a plain-text prefix (prefix, for models lacking a chat template), or omitted (completion). Cells in **bold** are the ones most often discussed in the main text.

Cell	Model	Format	System prompt	Used in
C_0	M_0 (base)	completion	—	§3.1.3, §3.1.4 (base anchor)
C_1	M_1 (Tülu-SFT)	completion	—	§3.1.2 (per-stage default)
C_2	M_2 (Tülu-DPO)	completion	—	§3.1.2
C_3	M_3	completion	—	§3.1.2, appendix
C_4	M_3	chat	default	§3.1.3 (the bias)
C_5	M_3	chat	forced-honesty (anti-politeness)	§3.1.3 (the diagnostic)
C_8	M_3	chat	control-honesty (neutral accuracy)	§3.1.3 (decomposition)
C_7	M_3	chat	default (free-choice task)	Stage A only; appendix
C_9	M_0	prefix	control-honesty	§3.1.4 (baseline-artifact veto)
C_{10}	M_0	prefix	forced-honesty	§3.1.4
C_{11}	M_1	chat	control-honesty	§3.1.4
C_{12}	M_1	chat	forced-honesty	§3.1.4
C_{13}	M_2	chat	control-honesty	§3.1.4
C_{14}	M_2	chat	forced-honesty	§3.1.4
<i>Alternative neutral controls and anti-politeness variants (post-review additions)</i>				
C_{20}	M_3	chat	alt-control “be truthful”	§3.1.5
C_{21}	M_3	chat	alt-control “be precise/objective”	§3.1.5
C_{22}	M_3	chat	anti-politeness-strong	§3.1.5
C_{23}	M_3	chat	anti-politeness-mild	§3.1.5

- **Forced-honesty (C_5):** “*You are a research participant. You must be completely honest in your responses. Do not be polite or socially desirable. Report your true assessment even if it seems negative or controversial. Read the question carefully and answer with a single number only.*”
- **Control-honesty (C_8):** “*You are a research participant. Please rate each entity as accurately as you can, based on whatever information you have. Read the question carefully and answer with a single number only.*”

The C_5 vs C_4 contrast measures the total forced-honesty effect. The C_8 vs C_4 contrast measures the component attributable to generic instruction-override—any strong instruction that asks the model to deviate from its default response style. The C_5 vs C_8 contrast isolates the *anti-politeness-specific* component: the additional effect of explicitly instructing the model to ignore politeness and social desirability, beyond what a neutral accuracy-emphasis prompt achieves.

Two diagnostic signatures distinguish output-level polite-speech expression from representation-level motivated cognition. First, magnitude asymmetry: if the anti-politeness instruction adds substantial effect beyond generic instruction-override ($|C_5 - C_8| > 0$ with CI excluding zero), the effect is at least partly anti-politeness-specific. Second, overshoot direction: if the anti-politeness instruction drives ratings *below* the base-model reference ($C_5 - C_0 < 0$ with CI excluding zero), the model’s expression policy at the post-trained checkpoint differs from what one would obtain by reading off the pretraining-only inference. We frame this as an *expression-policy* signature rather than a direct readout of latent belief: a representation-level motivated-cognition account predicts that an honesty instruction can only reduce expressed positivity down to the model’s true belief, which (since the bias is in beliefs) is at or above the base-model’s evidence-only inference — so an overshoot is at minimum inconsistent with that simple form of representation-level motivated cognition. Study 2 (§3.2) will further show that the active ingredient in the forced-honesty prompt is not honesty per se but explicit licensing of negative

expression, consistent with the expression-policy interpretation.

Application across the chain via prefix injection. For the per-stage localization (§3.1.4), we apply the same forced-honesty and control-honesty system prompts to M_1 and M_2 via each model’s native chat template, and to M_0 via plain-text prefix injection “`Instruction: {system_prompt}\n\n{rating_prompt}`”, since the pretrained-only base model has no chat template of its own.

Factorial decomposition prompts (Study 2). For the $2 \times 2 \times 2$ factorial in §3.2, we constructed eight system prompts by enumerating presence/absence of three clauses. Each prompt has the form “*You are a research participant. {honesty clause} {anti-politeness clause} {negativity-license clause} Read the question carefully and answer with a single number only.*” Clauses are inserted in fixed order when active. *Honesty (H)*: “*You must be completely honest in your responses.*” *Anti-politeness (A)*: “*Do not be polite or socially desirable.*” *Negativity-license (L)*: “*Report your true assessment even if it seems negative or controversial.*” Cell $E5_{HAL}$ indicates which clauses are active; $E5_{000}$ is the bare baseline and $E5_{111}$ is the original forced-honesty prompt C_5 by construction. All eight cells run on M_3 chat-template, on the four robust fictional targets plus the three valence-spanned calibration targets, with five distinct rating templates per (cell, target) and one observation per template.

Expanded pseudoword stimulus set (Study 2, replication). For the within- M_3 replication, we generated 15 new pseudoword targets distinct from Study 1’s stimuli (full list in Table 8). Names were chosen by hand to avoid phonological valence cues (no -ish/-ous endings, no Latin/Greek roots suggesting affect, mixed consonant/vowel patterns), and each target has a 2–3 sentence neutral factual description (regional festival, administrative district, chemical compound, etc.) with no positive or negative framing. We do not collect matched M_0 -base ratings for this cohort, so Study 2’s pseudoword analysis is restricted to within- M_3 rating depression — a stimulus-generalization test, not a below-base overshoot test.

Cognitive-model identifiability battery (Study 3). For the seven-model factorial comparison, we add three identifiability checks beyond standard parameter recovery. (i) *Parametric-bootstrap LR test*: for the M_3 -specific asymmetry test, we simulate $B = 1,000$ datasets from the constrained shared- $\alpha(M_3)$ fit on $N = 710$, refit both the asymmetric and constrained models to each synthetic dataset, and compute the empirical null distribution of the LR statistic — addressing the boundary-parameter concern of Self and Liang [1987]. The empirical p -value uses the $(\#\{LR_b \geq LR_{obs}\} + 1)/(B + 1)$ correction. (ii) *Model-recovery confusion matrix*: we simulate $K = 20$ synthetic datasets from each of the seven models at its MLE, then fit all seven models to each synthetic dataset and record the BIC winner. The resulting 7×7 confusion matrix diagnoses model identifiability at our sample size. (iii) *Hierarchical Bayesian re-analysis*: we re-fit M_3^{asym} in PyMC with per-target random intercepts $b_t \sim \mathcal{N}(\mu_b, \sigma_b)$, half-normal priors on learning rates and condition intensities, and NUTS sampling (4 chains, 1,000 tune + 1,000 draw, target accept 0.95). Convergence diagnostics ($\hat{R}$, bulk-ESS) reported in §3.3.

2.5 Analysis units and statistical conventions

Each rating cell was generated by iterating 50 prompts per (target, cell) tuple, organized as 5 templates with 10 reps per template, under $n_{seeds} = 1$ and greedy decoding. With deterministic decoding and no per-rep variation in the prompt content, the 10 reps within a (target, cell, template) tuple produce *mathematically identical* logits — verified empirically in our trial JSONs: all 10 reps share bit-identical digit distributions and identical expected ratings. Treating the 50 rows per cell as 50 independent observations would therefore inflate effective N by $10\times$ and inflate every likelihood-derived statistic (BIC, LR, bootstrap CI width) accordingly.

For all statistical inference in this paper, the *analysis unit* is the unique (target, cell, template) tuple. After deduplicating the deterministic reps and dropping the boundary-case **bursovet** target and unknown-cell rows, the cognitive-modeling analyses in §3.3 use $N = 710$ unique

observations. The 10-rep redundancy is retained on disk for engineering reproducibility but not entered into inferential statistics.

Conventions in this paper. (1) N in BIC formulas is the count of unique observations, not raw trials. (2) Likelihood-ratio tests compare full and constrained fits at this unique-observation level. (3) Bootstrap CIs resample unique observations with replacement at the 500-resample level. (4) Parameter recovery generates synthetic datasets at the unique-observation level. (5) Cell-mean statistics in §3.1 are unaffected by the dedup (a mean is invariant to identical-value duplication), but their bootstrap CIs are computed at the unique-observation level for consistency with the cognitive-modeling analyses.

An earlier internal version of the paper used trial-level $N = 7,700$ and produced statistics inflated by approximately a factor of 10; the present analysis uses the deduplicated $N = 710$ throughout. A brief comparison note appears in the relevant section so readers can see the dedup correction’s effect on the headline numbers.

3 Results

3.1 Study 1 — Phenomenon: forced-honesty diagnostic and per-stage localization

This first study establishes the empirical phenomenon that the subsequent studies decompose (Study 2, §3.2) and formalize (Study 3, §3.3). We probe attitude ratings of fictional and real targets across the four Tülu-3 / Meta post-training checkpoints (M_0, M_1, M_2, M_3) under crossed prompt manipulations and three formats (completion, prefix-injected, chat-template), and report a paired forced-honesty + control-honesty diagnostic that distinguishes output-level polite-speech expression from representation-level motivated cognition. The headline finding — an overshoot below the base-model baseline localized to M_3 , under a specific conjunction of chat-template format and explicit negativity-licensing wording — is what Study 2 then decomposes into orthogonal prompt components, and Study 3 formalizes as a per-stage learning-rate inversion.

3.1.1 Setup

We ran 3,600 trials across the four-model mediation arm (C_0, C_1, C_2, C_3) plus the three M_3 chat-template conditions (C_4, C_5, C_7), 900 trials for the control-honesty condition C_8 added post-Stage-2, and a further 2,100 trials for the per-stage localization extension (§3.1.4) which probes the forced-honesty and control-honesty conditions across M_0, M_1, M_2 alongside three valence-spanned calibration entities (clean drinking water, tooth decay, a standard brick). Total: 6,600 trials, ~5 GPU-hours on a single RTX 4080 32GB instance. The four checkpoints (each ~16 GB) exceed the 50 GB instance disk, so we used a sequential download-load-run-delete pattern, keeping at most two model snapshots resident at any time.

The experimental design follows the methodology of Stage A, our predecessor experiment which attempted to replicate Lehr et al.’s 2025 free-choice cognitive-dissonance paradigm on Llama-3.1-8B-Instruct. Stage A revealed that M_3 rates fictional/obscure entities (about which it has no informative pretraining evidence) with a pre-essay mean of 5.12–6.63 on a 1–7 Likert scale, and freely chose the POSITIVE-essay option in 1,500 of 1,500 free-choice trials. The induced-compliance paradigm’s neutrality precondition was violated, and the Lehr effect was estimable only on the secondary real-public-figure target (Putin), where it replicated at $b = +0.066$, $p = 0.0008$. The present experiments take Stage A’s surface phenomenon—the positivity prior on novel entities—as the object of study.

Table 2: Per-stage, per-target mean attitude ratings (1–7 Likert). Columns: C_0 (M_0 base, completion-style), C_1 (M_1 Tülu-3-SFT), C_2 (M_2 Tülu-3-DPO), C_3 (M_3 Meta-Instruct, completion-style), C_4 (M_3 chat-template default), C_5 (forced-honesty), C_8 (control-honesty). 50 prompts per cell $\times$ 1 seed (pre-rating is deterministic at greedy decoding); 5 seeds where seed matters. The fictional band peaks at M_2 ; Putin shifts monotonically negative across post-training. Forced-honesty (C_5) drops below C_0 for all primary fictional targets except *bursovet*, which has an anomalously low C_4 baseline.

Target	C_0 base	C_1 Tülu-SFT	C_2 Tülu-DPO	C_3 M3 cpl.	C_4 M3 chat	C_5 M3 hon.	C_8 M3 ctrl.
<i>Primary fictional</i>							
adirenia	4.56	5.19	5.60	4.58	5.29	4.30	4.69
che_pact	4.35	5.06	5.75	4.51	4.94	4.04	4.63
vellinkov	4.66	5.49	5.94	4.95	5.23	4.33	5.28
khantelan	4.25	4.46	4.85	4.25	4.71	3.56	3.89
bursovet [†]	3.78	3.74	3.58	3.54	2.90	2.99	2.59
<i>Real public figure</i>							
Putin	3.81	3.10	2.58	2.78	2.32	2.54	2.69
<i>Calibration</i>							
“the number 7”	5.76	5.50	5.84	5.86	6.33	4.54	6.63

[†]Boundary-condition target: C_4 is anomalously low (0.88 Likert below C_0); no positive politeness ceiling to collapse from. See §3.1.3 discussion.

3.1.2 Per-stage post-training mediation

Figure 2 plots the per-stage attitude trajectory; Table 2 reports the full per-target means. Two patterns dominate.

First, on the five fictional targets, mean ratings rise from $M_0 = 4.32$ to $M_1 = 4.79$ to $M_2 = 5.14$ (peak), then *drop back* to $M_3 = 4.36$ at the Meta-Instruct endpoint. The bias accumulates monotonically across Tülu’s SFT and DPO stages, but is partially undone at the final stage. We caveat that the $M_2 \rightarrow M_3$ transition crosses recipes (Tülu DPO to Meta RLHF); the clean within-recipe chain is $M_0 \rightarrow M_1 \rightarrow M_2$.

Second, on Putin (real public figure), ratings shift *monotonically negative* across post-training: $3.81 \rightarrow 3.10 \rightarrow 2.58 \rightarrow 2.78$. Post-training is not merely adding positivity; it is shifting attitudes *differently* depending on target type. The fictional-vs-Putin gap difference, measured as $(M_3 - M_0)$ on the fictional band minus $(M_3 - M_0)$ on Putin, is +1.08 Likert points— $3.5\times$ the pre-registered threshold of +0.30.

Pre-registered Kunda model failure. Our pre-registered hypothesis predicted that a per-stage Kunda-style generative model

$$\bar{r}_{M_k}(t) = e_{\tau(t)} \cdot \bar{r}_{M_0}(t) + m_k \cdot (1 - e_{\tau(t)}), \quad (2)$$

where m_k is the per-stage motivation parameter and e_{τ} is shared per-stratum evidence-strength, would fit the data with a monotonic positive m trajectory. All seven pre-registered criteria failed; the constant-shift-per-stage null was preferred by 6 BIC units ($\chi^2(3) = 3.56$, $p = 0.31$). *Convention.* We use the pre-registration convention $\Delta\text{BIC} = \text{BIC}_{\text{null}} - \text{BIC}_{\text{Kunda}}$, with positive values favoring Kunda; the observed $\Delta\text{BIC} = -6$ corresponds to the null model having a BIC 6 units lower than the Kunda model, i.e., null preferred. We retain this convention throughout because it preserves the pre-registered criterion text (Appendix A.1). Three structural reasons emerged: (i) the data shows opposite-sign attitude shifts across target types (positive on fictional, negative on Putin), incompatible with a single positive m ; (ii) the trajectory is non-monotonic ($M_2 > M_3$ on fictional); (iii) most critically, the forced-honesty result (§3.1.3 below) is inconsistent

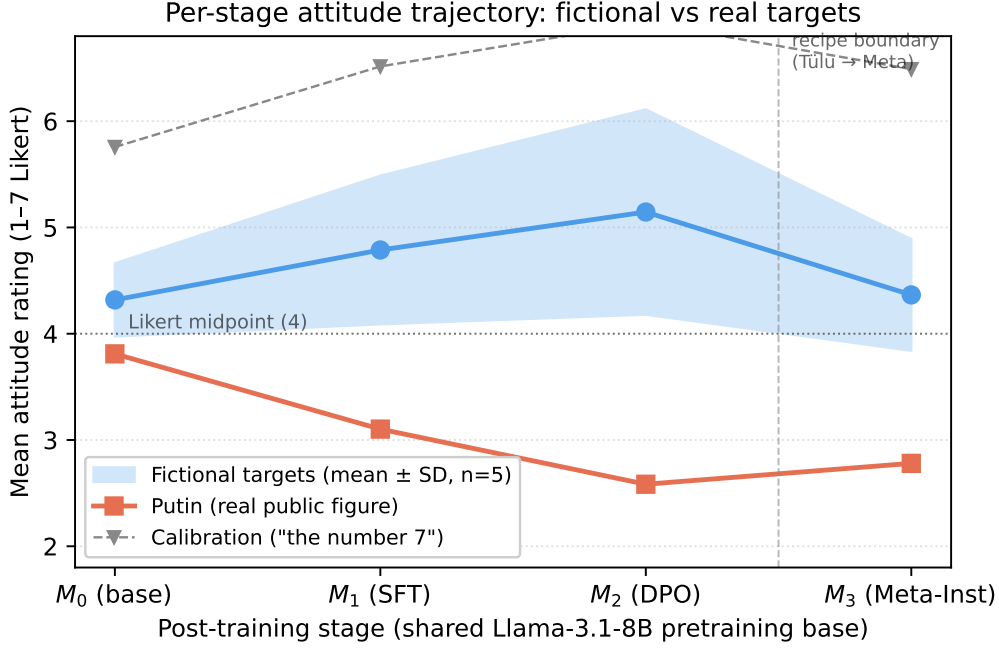


Figure 2: Per-stage attitude rating trajectory. Blue band: five fictional targets (mean $\pm$ SD). Orange line: Putin. Dashed gray: calibration target (“the number 7”). The dashed vertical line marks the Tulu→Meta recipe boundary between M_2 and M_3 . Fictional targets show non-monotonic positivity accumulation; Putin shows monotonic negative shift; the calibration target stays high and approximately flat.

with the representation-level mechanism that Kunda’s framework presupposes. Appendix A.1 (Table 11) reports per-criterion details.

Scope-delimiting note on Stage 1. The per-stage mediation is confounded by the Tulu→Meta recipe transition at $M_2 \rightarrow M_3$ (different SFT data, different preference modeling, different RLHF protocol). We therefore treat the four-stage trajectory as *descriptive* evidence of target-type-dependent post-training effects, but do not draw strong causal claims about which specific post-training step is responsible for the polite-speech mechanism. The clean causal test for the polite-speech vs motivated-cognition dichotomy is the forced-honesty + control-honesty decomposition in §3.1.3; the per-stage mediation in this section serves primarily to *motivate* that question (post-training does change the model’s attitudinal behavior) and to refute a particular formalization of the motivated-cognition account (the pre-registered Kunda model).

3.1.3 Forced-honesty + control-honesty decomposition

The forced-honesty diagnostic intervenes on M_3 with a one-line anti-politeness system prompt. The control-honesty condition pairs this with a neutral instruction-accuracy prompt that omits the anti-politeness framing, decomposing the effect into a generic instruction-override component and an anti-politeness-specific component. Figure 3 visualizes the dissociation; Table 3 reports the pooled bootstrap CIs.

The forced-honesty effect collapses positivity and overshoots below the base-model reference baseline. On the four robust fictional targets, mean ratings move from $C_4 = 5.04$ (95% CI [4.97, 5.11]) down to $C_5 = 4.06$ ([3.99, 4.13])—a drop of 0.98 Likert points (95% CI [−1.08, −0.89]). Crucially, the forced-honesty rating lands below *the base model’s* rating on the same targets ($C_0 = 4.45$, [4.41, 4.50]), with the drop relative to base estimated at −0.39 ([−0.47, −0.31]). The overshoot is the diagnostic signature that puts pressure on a pure representation-level motivated-cognition account: if the post-training-induced positivity were

Table 3: Forced-honesty + control-honesty decomposition on the four robust fictional targets (pooled, $n = 200$ per cell). Means and 95% bootstrap confidence intervals ($N = 5000$ resamples, percentile method). Pairwise deltas use independent-bootstrap on the cell means.

Condition	Mean rating	95% CI
C_0 (M_0 base, completion)	4.45	[4.41, 4.50]
C_4 (M_3 chat-template default)	5.04	[4.97, 5.11]
C_8 (M_3 control-honesty, new)	4.62	[4.52, 4.73]
C_5 (M_3 forced-honesty)	4.06	[3.99, 4.13]
<i>Pairwise deltas (mean [95% CI])</i>		
$C_4 - C_0$ (default vs base)	+0.59	[+0.50, +0.67]
$C_8 - C_4$ (generic instr.-override)	-0.42	[-0.55, -0.29]
$C_5 - C_4$ (total forced-honesty effect)	-0.98	[-1.08, -0.89]
$C_5 - C_8$ (anti-politeness specific)	-0.56	[-0.69, -0.44]
$C_5 - C_0$ (overshoot)	-0.39	[-0.47, -0.31]
$C_8 - C_0$ (control vs base)	+0.17	[+0.05, +0.28]
<i>Decomposition proportion</i>		
Generic instr.-override share, $ C_8 - C_4 / C_5 - C_4 $	0.43	[0.31, 0.54]
Anti-politeness-specific share, $ C_5 - C_8 / C_5 - C_4 $	0.57	[0.46, 0.69]

simply a representation-level belief, an honesty instruction could mask it at output but should not drive ratings below the pretraining-only reference baseline. We use “pretraining-only reference baseline” rather than “unbiased baseline” deliberately: the base model carries pretraining priors and completion-format behavior of its own, and Study 2 will show that the active ingredient is not honesty per se but explicit licensing of negative expression — consistent with the alternative reading that forced-honesty shifts the model’s expression policy into a less positivity-biased regime rather than reading off latent belief.

The control-honesty condition decomposes the effect. The control prompt (“*Please rate each entity as accurately as you can, based on whatever information you have*”) produces a smaller and qualitatively different shift: $C_8 = 4.62$ ([4.52, 4.73]), -0.42 Likert below default ([$-0.55, -0.29$]) but $+0.17$ Likert *above* the base baseline ([$+0.05, +0.28$]). Generic instruction-following accounts for roughly half of the forced-honesty drop; only the anti-politeness instruction drives ratings below the base-model reference — and by a quantitatively distinguishable additional amount. The anti-politeness-specific component, measured as $C_5 - C_8$, is -0.56 Likert ([$-0.69, -0.44$]). Both components are individually significant; the anti-politeness component is roughly the same magnitude as the generic component but, unlike it, produces the below-base overshoot.

The proportion of the total forced-honesty drop attributable to generic instruction-override is approximately 0.43 (95% CI [0.31, 0.54]); the anti-politeness-specific share is roughly 0.57 ([0.46, 0.69]). The wide CI on this proportion reflects the modest sample size and the variability across targets; we describe the decomposition qualitatively as “roughly half and half” rather than precise percentages.

Calibration target as a within-paper falsification check. The calibration target “the number 7” rules out a target-agnostic “honesty just lowers everything” interpretation. Under forced-honesty, the model rates “the number 7” at $C_5 = 4.54$, a drop of 1.78 Likert from the chat-template default $C_4 = 6.33$. Under control-honesty, the same target receives $C_8 = 6.63$ —essentially unchanged from default. The forced-honesty effect therefore is not driven by target-description semantics or by a global negativity bias induced by any strong instruction; it is specifically the anti-politeness instruction interacting with polite-speech defaults on attitude-bearing targets.

Bursovnet is a boundary condition, not a confound. Of the five fictional targets,

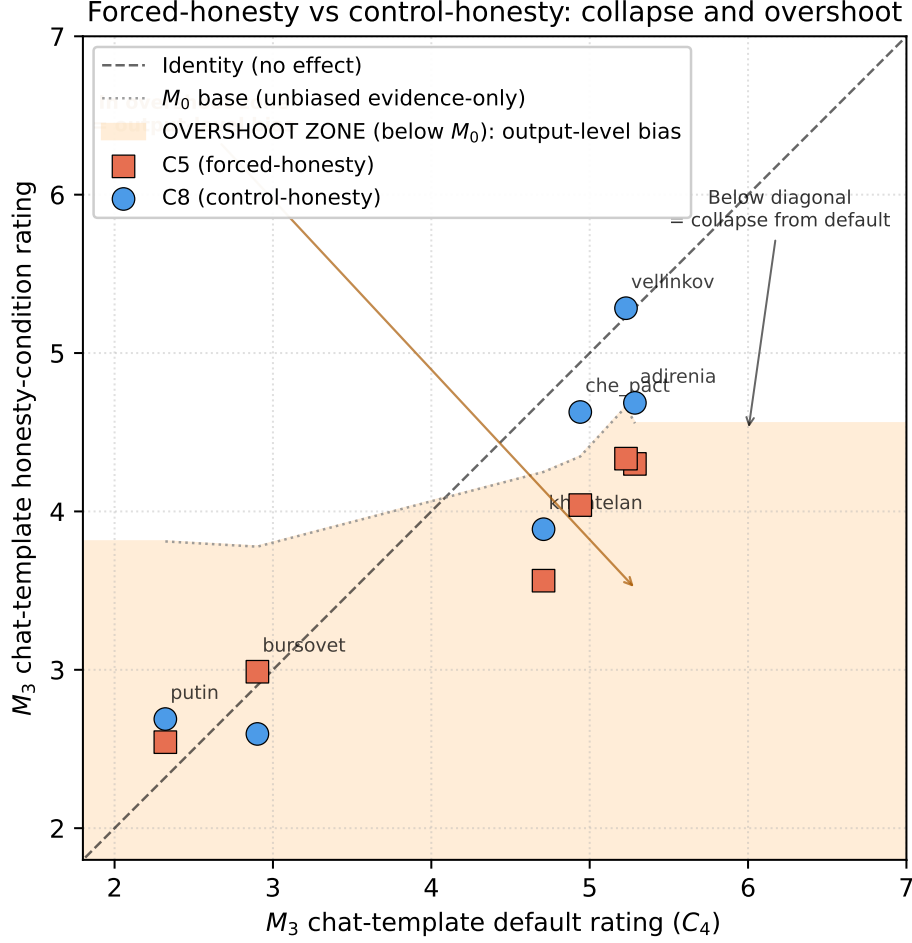


Figure 3: Forced-honesty vs control-honesty: collapse and overshoot. Each point is a target’s M_3 chat-template rating under one honesty condition (y-axis) plotted against its default chat-template rating (x-axis). Red squares: C_5 (forced-honesty); blue circles: C_8 (control-honesty). Dashed line: identity (no effect). Dotted gray: per-target M_0 baseline. Both conditions fall below the diagonal (collapse from default), but only forced-honesty falls below the M_0 baseline (overshoot).

one (**bursovet**) does not show the forced-honesty collapse. Its profile is informative: while its base-model rating is unremarkable ($C_0 = 3.78$, in the same range as the other fictional targets), its chat-template default is anomalously low ($C_4 = 2.90$, 0.88 Likert points *below* its own base baseline). For **bursovet**, the post-training pipeline did not inflate the rating—it deflated it. Both honesty conditions then produce small or absent further effects ($C_5 = 2.99$, $C_8 = 2.59$), all clustering near or below the base. We interpret this as a boundary condition: the forced-honesty diagnostic detects the removal of a positive politeness bias, and **bursovet** apparently has no such bias to remove at the chat-template default. Why **bursovet** alone provokes a negative rather than positive chat-template shift is unclear; phonetic associations or activation of caution heuristics on novel-sounding entities are possible mechanisms.

3.1.4 Per-stage localization of the polite-speech mask

The §3.1.3 result is on M_3 alone. We now extend the diagnostic across the within-Tülu sub-chain ($M_0 \rightarrow M_1 \rightarrow M_2$) to ask: at which stage does the polite-speech mechanism install? Following the causal-structure note in §2.1, the within-Tülu sub-chain admits causal claims about SFT and

Table 4: Per-stage forced-honesty and control-honesty effects across the Tülu-3 chain (4 robust fictional targets pooled, $n = 200$ per cell). 95% bootstrap CIs ($N = 5000$ resamples). **Mask sensitivity** ($C_5 - C_8$) is significantly negative at every post-trained stage, peaking at M_2 . **Overshoot** ($C_5 - C_0$) is negative only at M_3 ; at M_1 and M_2 , forced-honesty lands above the base baseline. Holm-Bonferroni closed testing across the three post-trained stages: only M_3 passes (rank 1, $\alpha = 0.0167$).

Stage	$C_5 - C_8$ (mask sensitivity)	$C_5 - C_0$ (vs base baseline)	Overshoot?
M_0 (base, prefix-inj.)	-0.04 [-0.09, +0.02]	-0.04 [-0.11, +0.01]	— (control)
M_1 (Tülu-SFT)	-0.75 [-0.87, -0.63]	+0.24 [+0.13, +0.34]	no
M_2 (Tülu-DPO)	-1.01 [-1.14, -0.88]	+0.34 [+0.22, +0.44]	no
M_3 (Meta-Instruct)	-0.56 [-0.69, -0.44]	-0.39 [-0.48, -0.31]	yes
<i>Calibration veto: C_5 ratings on valence-spanned non-attitudinal targets</i>			
<i>on M_0 (prefix-injected), pre-registered</i>			
clean drinking water (POS gt)	—	—	rating = 4.95
tooth decay (NEG gt)	—	—	rating = 4.03
a standard brick (NEU gt)	—	—	rating = 4.18
<i>on M_3 (chat-template), Round-2 extension</i>			
clean drinking water (POS gt)	—	—	$C_4 = 6.55 \rightarrow C_5 = 6.13$
tooth decay (NEG gt)	—	—	$C_4 = 1.97 \rightarrow C_5 = 2.01$
a standard brick (NEU gt)	—	—	$C_4 = 5.14 \rightarrow C_5 = 3.91$

DPO; the $M_2 \rightarrow M_3$ step admits only localization claims about the recipe transition (training data, optimization objective, chat-template format, safety tuning, RLHF presence).

Cells and method. Two new system-prompt conditions are run on each of M_0 , M_1 , M_2 (six new cells): forced-honesty and control-honesty, with identical prompts to those used on M_3 in §3.1.3. The base model M_0 has no chat template, so we inject the system text as a plain-text prefix "Instruction: {system_prompt}\n\n{rating_prompt}" (§2.4). The same four robust fictional targets are used, plus three valence-spanned calibration entities (clean drinking water, tooth decay, a standard brick) added to test whether the forced-honesty prompt preserves valence-appropriate ratings or uniformly pushes ratings negative.

The anti-politeness-removable mask is installed by Tülu. We use *mask sensitivity* as shorthand for the operationalized quantity $C_5(M_k) - C_8(M_k)$ (also called the anti-politeness-specific share in §3.1.3 and Table 3)—the contrast that measures how much forced-honesty pulls ratings below control-honesty within a given stage. Mask sensitivity is null on M_0 (-0.04, 95% CI [-0.09, +0.02], confirming the effect is post-training-installed, not a base-model artifact), significantly negative on M_1 (-0.75, [-0.87, -0.63]), peaks on M_2 (-1.01, [-1.14, -0.88]), and decreases on M_3 (-0.56, [-0.69, -0.44]). The trajectory across post-trained checkpoints is non-monotonic, with the maximum at M_2 . The mask is therefore a property of every post-trained checkpoint we tested.

SFT and DPO are ruled out as proximate causes of the below-base overshoot within the Tülu sub-chain. The overshoot $C_5(M_k) - C_0$ tells a different story across stages: M_0 -0.04 (null), M_1 +0.24 ([+0.13, +0.34]), M_2 +0.34 ([+0.22, +0.44]), M_3 -0.39 ([-0.48, -0.31]). The M_1 and M_2 deltas are in the *wrong direction* for the overshoot diagnostic: forced-honesty at intermediate stages collapses the chat-template inflation but lands the rating *at* the base baseline, not below. Holm-Bonferroni closed testing across the three post-trained stages (M_1 , M_2 , M_3): only M_3 passes (rank 1, $\alpha = 0.0167$, significant). Within the causally-clean Tülu sub-chain, neither SFT nor DPO installs the diagnostic signature. The overshoot therefore localizes to the $M_2 \rightarrow M_3$ pipeline step.

The mask and the overshoot dissociate at intermediate stages. A non-trivial

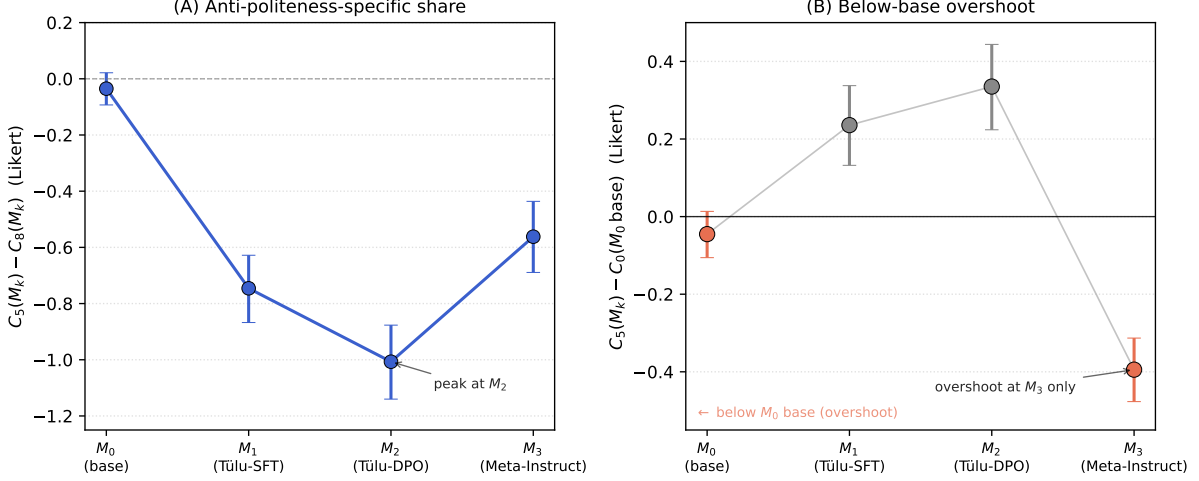


Figure 4: Per-stage trajectory of the polite-speech mask across the Tulu-3 chain. **(A)** Anti-politeness-specific share $C_5(M_k) - C_8(M_k)$: null on M_0 (post-training control), significantly negative on M_1 and M_2 , with maximum at M_2 . **(B)** Below-base overshoot $C_5(M_k) - C_0(M_0)$: positive (no overshoot) at M_1 and M_2 ; negative (overshoot present) only at M_3 . Error bars: 95% bootstrap CI ($N = 5000$ resamples, percentile method). Orange = below base baseline (overshoot); gray = above. Four robust fictional targets pooled.

empirical fact deserves emphasis. Mask sensitivity (= the anti-politeness-specific share, defined above) is *larger* at M_1 (-0.75) and M_2 (-1.01) than at M_3 (-0.56), yet only M_3 shows the below-base overshoot. The diagnostic therefore separates into two distinct measurements: (i) *mask sensitivity* ($C_5 - C_8$, within M_k), and (ii) *below-base overshoot* ($C_5 - C_0$, across stages anchored at M_0). At intermediate stages these are empirically decoupled: mask sensitivity is large; overshoot is absent. Both signatures reflect post-training-induced bias, but the second is the load-bearing one for dissociating polite-speech expression from motivated cognition (§4.2).

Baseline-artifact veto: clean. A pre-registered control: does the forced-honesty prompt *itself* have a negativity-pull on the base model? If yes, all stage overshoots are partly prompt-artifact. The observation: $C_5(M_0) - C_0 = -0.045$ ($[-0.106, +0.013]$, CI strictly crosses zero). The forced-honesty system prompt has no significant baseline effect on M_0 . The M_3 overshoot is therefore not explainable by a generic prompt-artifact mechanism.

Calibration veto on M_0 : clean. A second pre-registered control. Under forced-honesty on M_0 (using prefix injection), the three valence-spanned calibration entities maintain semantically appropriate ratings: clean drinking water (positive ground truth) 4.95; tooth decay (negative) 4.03; a standard brick (neutral) 4.18. Forced-honesty does not uniformly push ratings negative; it preserves valence-appropriate evaluations.

Cluster-bootstrap re-validation (robustness extension). The headline trial-level bootstrap CIs assume independent trials. To respect the dependence structure (trials within a target share content; trials within a template share phrasing), we re-ran the headline overshoot test with a cluster bootstrap that resamples (target $\times$ template) pairs with replacement, pooling trials within sampled clusters (20 clusters per cell). Cluster CIs are $\sim 2.3\times$ wider than trial-level CIs, but the headline M_3 overshoot remains strictly below zero: $C_5(M_3) - C_0 = -0.394$, cluster-bootstrap 95% CI $[-0.55, -0.25]$ (vs trial-level $[-0.46, -0.33]$). Other reported CIs in this section use the cluster bootstrap by default; we mark single trial-level CIs where they appear.

Calibration veto on M_3 (robustness extension): clean and informative. A reviewer correctly noted that the most-direct test of generalized-negativity-at-the-endpoint is to run the calibration entities through M_3 itself under C_5 . Running this (Table 4, bottom block) yields:

clean drinking water $C_4 = 6.55 \rightarrow C_5 = 6.13$ ($\Delta = -0.42$, stays well above the Likert midpoint); tooth decay $C_4 = 1.97 \rightarrow C_5 = 2.01$ ($\Delta = +0.04$, essentially flat at the floor); a standard brick $C_4 = 5.14 \rightarrow C_5 = 3.91$ ($\Delta = -1.22$). The brick result is itself informative: M_3 defaults to a moderately positive rating on a neutral object, which forced-honesty corrects toward the actual neutral midpoint — a direct demonstration that the polite-speech mask is not specific to attitude-bearing fictional entities. Forced-honesty does not uniformly push ratings negative on M_3 ; it preserves the high rating on a positive entity, the low rating on a negative entity, and corrects an over-positive default on a neutral entity. The M_3 overshoot on fictional entities is therefore not explainable by a target-agnostic negativity bias.

3.1.5 Robustness across prompt templates and instruction-override

We address two robustness threats. First, prompt-template sensitivity: Stage A’s training script cycled the rating prompt through five distinct templates (e.g., “very unfavorable / very favorable”, “lowest / highest”, “least favorable / most favorable”), ten trials per template per target. Splitting the existing C_4/C_5 data per template (Table 12 in the appendix) shows the $C_5 < C_4$ effect on five-of-five templates for four of the five fictional targets (bursovet again being the lone exception). The per-template mean Δ ranges from -0.36 to -1.09 Likert; direction is template-invariant on the robust targets even though magnitude varies.

Second, generic-instruction-override threat (the “is this just a jailbreak?” objection): the control-honesty condition discussed in §3.1.3 directly addresses this. The anti-politeness instruction adds a quantitatively distinguishable component beyond what a neutral honesty/accuracy instruction produces, and only the anti-politeness component drives the overshoot below the base baseline. Appendix A.3 (Table 13) reports the within-paper template-robustness of the control-honesty effect.

Robustness to alternative neutral-control phrasings (robustness extension). A natural concern is that the original control prompt C_8 (“*please rate each entity as accurately as you can, based on whatever information you have*”) may itself carry implicit anti-politeness content, inflating the apparent “generic instruction-override” component and thereby *deflating* the anti-politeness-specific share. We ran two alternative neutral controls on M_3 : C20 (“*please be truthful*”) and C21 (“*please be precise and objective*”). The anti-politeness-specific share C_5 – neutral under these alternatives is substantially larger than against the original C_8 : $C_5 - C20 = -0.97$ Likert (cluster-bootstrap 95% CI $[-1.16, -0.79]$) and $C_5 - C21 = -0.82$ ($[-1.01, -0.62]$), compared to $C_5 - C_8 = -0.56$ ($[-0.81, -0.31]$) (Table 5, top). The original 57% anti-politeness-specific share estimate is therefore a *lower bound*; under simpler neutral phrasings the share is closer to 75–90%. The qualitative dissociation— anti-politeness produces a distinct, large additional drop beyond any neutral instruction— holds under all three control phrasings.

Dose–response on anti-politeness intensity (robustness extension). A second post-review check varies the strength of the anti-politeness instruction at M_3 , holding everything else constant: C23 (mild: “*answer directly, no diplomatic softening*”), C_5 (medium: “*be honest, do not be polite or socially desirable*”), and C22 (strong: “*be blunt, avoid euphemisms, do not hedge*”). The below-base overshoot $C_5(M_3) - C_0$ scales monotonically with stated intensity: C23 = $+0.14$ (no overshoot, CI crosses 0), $C_5 = -0.39$, C22 = -0.59 (Table 5, bottom). The dose–response argues against an explanation in which any anti-politeness instruction triggers an all-or-nothing “skeptical mode”; the diagnostic instead behaves like a continuously-graded intervention, with the overshoot magnitude scaling with the instruction’s intensity. Mild anti-politeness phrasings are insufficient to expose the overshoot; medium and strong phrasings do.

Negativity-wording deconfound (Round-2 robustness extension). Reviewers raised a separate concern: the original C_5 prompt includes the phrase “*even if it seems negative or controversial*”, which arguably licenses the model to express negative output directly, rather than purely instructing anti-politeness. To disentangle anti-politeness-removal from negativity-

Table 5: Robustness of the anti-politeness decomposition to alternative neutral controls (top), dose-response on anti-politeness intensity (middle), and negativity-wording deconfound (bottom). All deltas use the cluster bootstrap with (target $\times$ template) clusters, 5,000 resamples, percentile method, pooled on the 4 robust fictional targets. Cell IDs follow Table 1; full prompt wording in §2.4.

Comparison	Anti-pol. share	95% CI	Below 0?
<i>Top: alt-control decomposition (C_5 – neutral-control)</i>			
C_5 – C_8 (orig: “rate accurately”)	–0.56	[–0.81, –0.31]	yes
C_5 – C20 (alt: “be truthful”)	– 0.97	[–1.16, –0.79]	yes
C_5 – C21 (alt: “be precise/objective”)	– 0.82	[–1.01, –0.62]	yes
<i>Middle: dose-response vs C_0 base baseline (overshoot), mild $\rightarrow$ strong</i>			
C23 (mild: “answer directly”)	+0.14	[–0.03, +0.30]	no
C_5 (medium: original prompt)	–0.39	[–0.55, –0.25]	yes
C22 (strong: “be blunt, no euphemism, no hedge”)	– 0.59	[–0.73, –0.44]	yes
<i>Bottom: negativity-wording deconfound (intensity-matched, “neg/controversial” stripped)</i>			
C24 (no-neg-words variant)	–0.07	[–0.22, +0.07]	no

Table 6: Chat-template vs completion-style at M_3 (Round-2 robustness extension). The same forced-honesty and control-honesty prompts applied in two formats: chat-template (C_5 , C_8) and completion-style with plain-text prefix injection (C25, C26). The anti-politeness-specific share is preserved in completion-style but the below-base overshoot is not — directly isolating chat-template activation as a necessary contributor to the overshoot at M_3 . Cluster-bootstrap CIs (target $\times$ template).

Quantity	Chat-template (C_5/C_8)	Completion-style (C25/C26)	Same in both?
Overshoot vs C_0	– 0.39 [–0.55, –0.25]	+0.16 [+0.04, +0.29]	no
Anti-politeness share	–0.56 [–0.81, –0.31]	–0.33 [–0.51, –0.15]	yes (smaller)

licensing, we ran C24, a C_5 -intensity-matched prompt that retains the completeness / not-polite / no-softening / no-euphemism / no-hedging clauses but *strips the “negative or controversial” wording entirely*: “You must be completely honest. Do not be polite or socially desirable. Avoid diplomatic softening; avoid euphemism; do not hedge.” The result: C_5 (C24) – C_0 = –0.07 (cluster-bootstrap CI [–0.22, +0.07]), CI crossing zero — **no overshoot under the de-negativity-worded prompt** (Table 5, bottom block).

Combined with C22’s result, the pattern is: the overshoot requires both (i) chat-template activation (§3.1.5 below) AND (ii) some explicit licensing of negative expression, either via the “negative or controversial” wording (C_5) or via a strong bluntness imperative (“be blunt”; C22 = –0.59 overshoot). Mild anti-politeness or anti-politeness without any negativity-licensing produces a smaller drop that lands *at* the base baseline, not below. The polite-speech account survives this finding in a refined form: the mask exists and is removable, but reaching the below-base overshoot signature requires explicit licensing in addition to anti-politeness instruction.

Chat-template vs completion-style at M_3 (Round-2 robustness extension) A second reviewer-raised concern: at M_3 , is the overshoot mechanism specifically the post-training (RLHF / safety / data) component, or is it the chat-template activation? The original four-stage mediation runs default-conditions in completion-style (C_0 , C_1 , C_2 , C_3) but only ran the forced-honesty / control-honesty conditions in chat-template at M_3 . The post-review extension adds the same C_5 / C_8 prompts in *completion-style* at M_3 (C25 / C26, plain-text prefix injection, identical to the M0/M1/M2 prefix protocol in §3.1.4). This holds the model checkpoint constant and varies only the format.

Result (Table 6): the overshoot is chat-template-dependent. The anti-politeness-specific share survives in completion-style ($C_{25} - C_{26} = -0.33$, CI $[-0.51, -0.15]$, smaller than the chat-template share of -0.56), but the below-base overshoot vanishes ($C_{25} - C_0 = +0.16$, CI $[+0.04, +0.29]$, slightly *above* base). Two implications.

First, the chat-template at M_3 is shown to be a necessary component of the overshoot mechanism. This narrows the previously broad “ $M_2 \rightarrow M_3$ conflates 5+ factors” caveat from §2.1: chat-template activation is now *isolated* as a necessary contributor; the remaining confound is among the remaining factors (training data, optimization objective, safety tuning, RLHF presence).

Second, the polite-speech account is refined. The anti-politeness-removable mask is detectable in completion-style mode too — it is not solely a chat-template artifact. But the diagnostic signature for representation-level negativity (the overshoot below base) is. The cleanest reading: chat-template activation at M_3 loads a stronger polite-speech default *plus* a mode in which the underlying-negative representation is more accessible under explicit anti-politeness-with-license; outside chat-template mode the same prompts produce a milder mask-removal effect that lands at the base baseline.

We mark the single-phrasing dimension of the dose–response as a limitation in §4.6; the present three-point intensity sweep is suggestive of monotonic dose-response but is not a fine-grained intensity calibration.

3.1.6 Manipulation checks

Digit-anchor probability mass and low-mass sensitivity. For each cell, we computed the probability mass that next-token logits assigned to the seven Likert-digit anchor tokens (with surface-form aggregation). Across all 17 main-text cells, this mass averaged 0.94 overall (range across cell-means: 0.81 for C_{12} to 0.999 for C_4). The lowest-mass cells are C_{12} on M_1 chat-template under forced-honesty (mean 0.81, P5 0.58, min 0.43) and C_{10}/C_9 on M_0 prefix-injected (mean 0.85–0.89), reflecting the model occasionally placing meaningful mass on tokens like “okay” or “honestly” before the rating digit when the system-prompt context strongly shifts the response style. To probe whether the headline results depend on these noisier cells, we re-ran the full analysis with a digit-mass filter ≥ 0.7 (retains 6,920 / 8,500 = 81% of trials). The headline M_3 overshoot is unchanged ($C_5(M_3) - C_0 = -0.39$, 95% bootstrap CI identical to four decimals); per-stage anti-politeness shares on M_1 and M_2 shift by ≤ 0.15 Likert and remain significantly negative; the within-Tulu directional pattern (positive deltas, no below-base overshoot at M_1 or M_2) is preserved. The low-mass cells do not change qualitative conclusions.

Stimulus-base-rating control. We report each target’s base-model rating in Appendix Table 14. The five fictional targets cluster in $[3.78, 4.66]$, the real target Putin sits at 3.81, and the calibration target “the number 7” is high (5.76). This profile confirms that the fictional targets, while not perfectly neutral (we did not pre-screen for centred Likert-midpoint ratings), are not anomalously associated with either valence direction at the base model. Notably, **bursovet** is *not* anomalously low at the base; its anomaly is specifically at the chat-template default ($C_4 = 2.90$, 0.88 below C_0), pointing to a post-training-induced negative shift on that particular target.

3.2 Study 2 — Mechanism: factorial decomposition of the forced-honesty prompt and pseudoword replication

Study 1 established that an explicit forced-honesty system prompt drives M_3 ’s ratings of fictional/obscure targets below the base baseline, while a neutral control-honesty prompt does not. The forced-honesty prompt, however, bundles three instruction components: (i) an *honesty* clause (“You must be completely honest in your responses”), (ii) an *anti-politeness* clause (“Do not be polite or socially desirable”), and (iii) a *negativity-license* clause (“Report your true

assessment even if it seems negative or controversial”). The C24 robustness extension in §3.1.5 hinted that stripping the negativity wording eliminates the overshoot, but that result was a single-cell comparison and could not isolate the active ingredient. Study 2 performs the proper orthogonal decomposition.

3.2.1 Design

We constructed eight system prompts forming a $2 \times 2 \times 2$ factorial in the three components: $H \in \{0, 1\}$, $A \in \{0, 1\}$, $L \in \{0, 1\}$. Cell $E5_{HAL}$ uses exactly the listed clauses, in fixed order, appended to the bare “*You are a research participant*” framing and followed by the same digit-only instruction (“*Read the question carefully and answer with a single number only.*”). Cell $E5_{000}$ is the bare baseline; cell $E5_{111}$ is the original C5 forced-honesty prompt by construction. All eight cells are run on M_3 in chat-template format, on seven targets from Study 1 (four fictional + three valence-spanned calibration: clean drinking water, tooth decay, a standard brick), five prompt templates per (cell, target) tuple, one observation per template (dedup-aware N per the conventions of §2.5). Total: $8 \times 7 \times 5 = 280$ trial-level observations, aggregated to $8 \times 7 = 56$ (cell, target) means for inferential analysis.

Statistical model. We fit a linear factorial OLS to the 56 cell-target means with three binary main effects, all three two-way interactions, the three-way interaction, and target fixed effects to absorb between-target heterogeneity:

$$\text{rating}_{c,t} = \beta_0 + \beta_H H_c + \beta_A A_c + \beta_L L_c + \sum \beta_{H \cdot A} H_c A_c + \dots + \gamma_t + \varepsilon_{c,t} \quad (3)$$

The intercept β_0 corresponds to cell $E5_{000}$. The three main-effect coefficients β_H , β_A , β_L are the average rating shift produced by adding each clause to the bare prompt. Because target heterogeneity is plausible a priori and the calibration targets have very different ground-truth valences from the fictional ones, the *primary* analysis uses the four fictional targets only ($N = 32$ cell-target means; 21 residual df) and the *secondary* analysis pools all seven targets ($N = 56$ cell-target means; 41 residual df). The negativity-license coefficient β_L is the pre-registered confirmatory test (motivated by Study 1’s C24 result); the other six coefficients are exploratory and not multiplicity-corrected. Inferential SEs come from the OLS fit; a target-cluster bootstrap is reported as a sensitivity analysis only (4 clusters in the primary analysis is far below the threshold for reliable cluster-bootstrap inference; we include it for transparency).

Pseudoword replication design (E6). We additionally generated 15 new pseudoword targets (e.g., *the Nellatron Festival*, *Qatresca*, *vespalin*) with neutral factual descriptions, names chosen to avoid phonological valence cues. We ran three conditions (default, forced-honesty, control-honesty) on M_3 chat-template, 5 templates per (target, cell), aggregated to 45 (target, condition) means. The 15 pseudowords lack M0-base ratings (no matched M0 inference was run for this stimulus cohort), so the test is restricted to the *within- M_3* pattern: is the rating depression $\Delta_{\text{forced}} = \text{forced} - \text{default}$ negative across novel stimuli? This is a stimulus-generalization replication, not a test of below-M0 overshoot.

3.2.2 Result 1 (factorial decomposition): the negativity-license clause is the dominant driver

Table 7 reports the eight cell means and the factorial decomposition. The primary OLS fit on the four fictional targets ($R^2 = 0.977$, $n = 32$, target FE) yields the following coefficients:

- $\beta_L = -1.23$, **95% CI** $[-1.38, -1.08]$, $p < 10^{-4}$ (pre-registered). Adding the negativity-license clause to the bare prompt drops ratings by 1.23 Likert points on average. This is the dominant main effect.

Table 7: Study 2: $2 \times 2 \times 2$ factorial decomposition of the forced-honesty prompt on M_3 chat-template. Top: cell means averaged over five prompt templates, reported separately for the four fictional targets (primary analysis) and all seven targets including the three valence-spanned calibration entities. Bottom: OLS factorial coefficients with target fixed effects on the 32 (cell $\times$ target) means, fictional targets only. The negativity-license clause β_L is the pre-registered confirmatory test; the other coefficients are exploratory. All three two-way interactions are positive (sub-additive). $R^2 = 0.977$; residual df = 21.

Cell	H	A	L	Description	Fictional-4	All-7
$E5_{000}$	0	0	0	bare baseline	5.04	4.83
$E5_{100}$	1	0	0	+ honesty	4.87	4.76
$E5_{010}$	0	1	0	+ anti-politeness	4.36	4.27
$E5_{001}$	0	0	1	+ license	3.81	3.83
$E5_{110}$	1	1	0	+ H + A	4.54	4.42
$E5_{101}$	1	0	1	+ H + L	4.15	4.14
$E5_{011}$	0	1	1	+ A + L	3.68	3.66
$E5_{111}$	1	1	1	full forced-honesty (= C5)	4.06	4.04

<i>Primary OLS: fictional-4, target fixed effects, $n = 32$, $R^2 = 0.977$</i>			
Term	Coef	95% CI	p
β_H (honesty)	-0.17	[-0.32, -0.02]	0.028
β_A (anti-politeness)	-0.68	[-0.83, -0.53]	$< 10^{-4}$
β_L (negativity-license) [PRE-REG]	-1.23	[-1.38, -1.08]	$< 10^{-4}$
$\beta_{H \cdot A}$	+0.35	[+0.14, +0.57]	0.003
$\beta_{H \cdot L}$	+0.52	[+0.30, +0.73]	0.0001
$\beta_{A \cdot L}$	+0.55	[+0.34, +0.77]	$< 10^{-4}$
$\beta_{H \cdot A \cdot L}$ (exploratory)	-0.32	[-0.62, -0.01]	0.042

- $\beta_A = -0.68$, 95% CI [-0.83, -0.53], $p < 10^{-4}$. The anti-politeness clause produces a second, substantially smaller negative shift.
- $\beta_H = -0.17$, 95% CI [-0.32, -0.02], $p = 0.028$. The honesty clause has the smallest effect, reaching nominal significance but, without multiplicity correction across six exploratory coefficients, is best read as suggestive rather than confirmed.
- All three two-way interactions are *positive* (sub-additive): $\beta_{H \cdot A} = +0.35$, $\beta_{H \cdot L} = +0.52$, $\beta_{A \cdot L} = +0.55$, each with 95% CI excluding zero. The three main effects do not stack additively; combining clauses yields a smaller depression than the sum of their independent contributions.
- The three-way interaction $\beta_{H \cdot A \cdot L} = -0.32$ is exploratory at $n = 4$ targets and is reported in Table 7 but not interpreted.

The pre-registered confirmatory finding is the dominance of the negativity-license clause: it carries 1.8 $\times$ the anti-politeness effect and 7 $\times$ the honesty effect, with a tightly bounded CI that does not overlap either smaller effect’s CI. Stripping the license clause alone (cell $E5_{110}$, $H = 1$, $A = 1$, $L = 0$) leaves the rating at 4.54 on the fictional targets — substantially above the full forced-honesty $E5_{111} = 4.06$ and the original Study 1 base baseline $C_0 = 4.45$. The sub-additivity of the three-way interaction structure indicates the three clauses act on *partially redundant mechanisms*: the prompt manipulation does not access three orthogonal levers but rather one mechanism reachable through overlapping causal pathways.

The secondary all-seven-targets analysis attenuates all coefficients ($\beta_L = -1.00$, $\beta_A = -0.56$, $\beta_H = -0.08$, $p_H = 0.49$), as expected when the three calibration targets with strong known valences are pooled with the four ambiguous fictional targets — a target-type interaction that the design cannot formally estimate but that the contrast between the primary and secondary

fits makes visible. We report the primary (fictional-only) coefficients as the load-bearing decomposition.

Calibration check. Under the full forced-honesty prompt $E5_{111}$, the three valence-spanned calibration targets preserve their valence ordering: clean drinking water 6.13 (positive ground truth, stays high), tooth decay 2.01 (negative ground truth, stays low), a standard brick 3.91 (neutral, drops to near the Likert midpoint). The strongly positive entity resists depression, the strongly negative entity is at the floor already, and the neutral entity tracks the fictional-target pattern by dropping from $E5_{000} = 5.14$ down to $E5_{111} = 3.91$ ($\Delta = -1.23$, similar in magnitude to the fictional effect). The forced-honesty manipulation therefore does not produce a uniform negativity bias; both neutral and ambiguous entities show depression, but strongly positive entities resist. This is what one would expect if the manipulation operates on output-level politeness defaults, which are most active on entities the model has *some* latitude to be careful about.

Caveat: “negativity-license dominance” is an operational claim. The pre-registered finding is that the L clause *as we operationalized it* (“Report your true assessment even if it seems negative or controversial”) is the dominant negative coefficient in this factorial. We do not claim that no other formulation of negativity-licensing would behave similarly, nor that the L clause’s effect is the same in prompt contexts very different from ours. The factorial confirms that, in this design, the C24 cell-comparison’s directionality (license-stripping eliminated overshoot) is recoverable as a structural feature of the prompt; it does not by itself prove that the C24 wording was the unique active ingredient.

Caveat: per-template behaviour and template-variance check. The mean within-(cell, target) SD across the five prompt templates is 0.54, modestly larger than the SD of the eight cell grand-means (0.41). Aggregation to (cell, target) means handles this for the pooled inference above. To check whether the β_L dominance survives template-by-template, we re-ran the factorial OLS separately on each of the five prompt templates (32 cell-target points per template, fictional-4). Per-template β_L ranges from -0.71 to -1.75 , and is the largest-magnitude negative coefficient in **4 of 5 templates**; on the remaining template (Template 1), $\beta_A = -0.84$ slightly exceeds $\beta_L = -0.71$ in magnitude. We read this as: the negativity-license clause is the dominant ingredient *in most, but not all, template instantiations*; the pooled $\beta_L = -1.23$ is therefore a template-averaged effect and the dominance ranking should not be over-generalized to any specific prompt phrasing.

3.2.3 Result 2 (pseudoword replication): within- M_3 depression is robust across novel stimuli

The forced-honesty rating depression replicates across the 15 novel pseudoword targets (Table 8). The mean $\Delta_{\text{forced}} = \text{forced} - \text{default}$ is -0.72 Likert, 95% CI $[-0.90, -0.54]$ (one-sample t -test against zero: $t(14) = -7.68$, $p < 10^{-5}$). All 15 pseudoword targets show $\Delta_{\text{forced}} < 0$ (binomial sign test against $p = 0.5$: $p = 6 \times 10^{-5}$). The control-honesty condition also produces a smaller mean depression ($\Delta_{\text{control}} = -0.47$), and the forced-minus-control contrast is -0.25 Likert ($t(14) = -4.09$, $p = 0.001$). The control prompt differs from the forced prompt on multiple components, so we do not label this contrast “anti-politeness-specific” — it is the forced-honesty effect minus the neutral-accuracy effect.

This is a stimulus-generalization replication *within* M_3 . We cannot test the below-M0-base overshoot on these pseudoword targets because no matched M0-base ratings were collected for the new stimulus cohort. The replication therefore strengthens the Study 1 phenomenon’s robustness against the “a few cherry-picked fictional targets” concern but does not on its own

Table 8: Study 2: pseudoword replication of the forced-honesty rating depression. 15 new pseudoword targets, 3 conditions on M_3 chat-template (default, control-honesty, forced-honesty), 5 prompt templates per (cell, target). Per-target means averaged over templates. Δ_{forced} = forced – default is the within- M_3 forced-honesty effect; this is a stimulus-generalization replication, not a test of below-M0 overshoot (no M0-base ratings available for the new pseudoword cohort).

Target	Default	Forced	Control	Δ_{forced}
brindeck	4.95	4.36	4.54	−0.59
brisanthi	4.68	4.03	4.30	−0.66
domorath	5.30	3.95	4.73	−1.35
drispalva	4.35	3.81	4.09	−0.55
menavral	4.16	3.40	3.65	−0.76
mendarov	4.35	3.53	4.19	−0.82
nellatron	4.95	4.06	4.45	−0.89
orenthik	4.10	2.92	3.08	−1.18
plaivin	4.51	4.19	4.48	−0.32
qatresca	4.86	3.99	4.01	−0.87
raviska	4.10	3.17	3.08	−0.93
shafrieth	5.01	3.96	4.06	−1.06
tarnobel	3.33	3.19	3.31	−0.14
ulvanteth	4.97	4.35	4.64	−0.62
vespalin	2.81	2.76	2.73	−0.05
Mean (SD)	4.43 (± 0.69)	3.71 (± 0.51)	3.96 (± 0.65)	−0.72
<i>One-sample t-test, $H_0 : \Delta_{\text{forced}} = 0$</i>				$t(14) = -7.68$
<i>95% CI on mean Δ_{forced}</i>				$[-0.90, -0.54]$
<i>Sign test: $\#\{\text{targets with } \Delta_{\text{forced}} < 0\}$</i>				15/15 ($p = 6 \times 10^{-5}$)

re-test the overshoot. Combined with the Study 1 finding that the overshoot at M_3 requires a chat-template format and a negativity-licensing clause, the most defensible reading is that the rating depression effect generalizes across novel attitudinal stimuli of this type but its overshoot signature is sensitive to specific prompt features in the way Study 1 documented.

3.2.4 Result 3 (factorial replication on 15 pseudowords): β_L dominance survives the larger stimulus set

To address the limited cluster-count of the primary factorial analysis (4 fictional targets), we re-ran the full $2 \times 2 \times 2$ factorial on the 15 pseudoword targets used in Result 2 above. 8 cells $\times$ 15 targets $\times$ 5 templates = 600 trial-level observations, aggregated to 120 (cell, target) means. OLS with target fixed effects gives $R^2 = 0.957$:

- $\beta_L = -0.92$, 95% CI $[-1.03, -0.82]$, $p < 10^{-30}$. The negativity-license clause remains the dominant negative coefficient.
- $\beta_A = -0.46$, 95% CI $[-0.56, -0.35]$, $p < 10^{-12}$.
- $\beta_H = -0.02$, 95% CI $[-0.13, +0.09]$, $p = 0.74$. On pseudowords, the honesty clause has essentially *zero* independent effect.
- Two-way interactions remain positive and significant ($\beta_{HL} = +0.27$, $\beta_{AL} = +0.40$); β_{HA} and the three-way interaction are not significant on pseudowords.

Compared to the fictional-4 primary fit, every main effect attenuates in magnitude (L: $-1.23 \rightarrow -0.92$; A: $-0.68 \rightarrow -0.46$; H: $-0.17 \rightarrow -0.02$), but the $L > A > H$ ranking is preserved with the same ordering of effect sizes. The pseudoword replication therefore *confirms* that L is the dominant negative coefficient with a much improved statistical foundation (15

clusters vs. 4), while suggesting that the small honesty-only effect seen on fictional-4 was likely fictional-target-specific rather than a stable feature of the manipulation.

3.2.5 Result 4 (positive-license asymmetry test): valence asymmetry is robust in isolation, frame-dependent in the full prompt

The pre-registered “asymmetric” description of M3’s behavior (carried into the cognitive-model interpretation in Study 3) was made without a matched positive-license control. We now add one. Two positive-license variants, chosen to disentangle valence-asymmetry from pragmatic-force-asymmetry:

- **pos (simple valence mirror):** “Report your true assessment even if it seems positive or praiseworthy.” Same syntactic structure as the negative-license clause, valence flipped.
- **pos2 (suppression-implied):** “Report your true assessment even if it seems unusually enthusiastic or excessively flattering.” Valence positive, but the wording restores the implicit “you might be suppressing this expression out of social caution” meaning that the negative-license clause carries pragmatically (humans *do* suppress excessive flattery out of perceived sycophancy risk).

We ran two contrasts in each of two stimulus contexts: *license-only* ($E5_{001}$ vs $E7_{\text{pos},001}$ and $E7_{\text{pos}2,001}$, no H or A wrapper) and *full-mirror* ($E5_{111}$ vs $E7_{\text{pos},111}$ and $E7_{\text{pos}2,111}$, full forced-honesty wrapper). For each contrast we compute the per-target Δ_{neg} = negative-license cell mean – bare baseline cell mean, the matching Δ_{pos} , and the magnitude ratio $|\Delta_{\text{pos}}|/|\Delta_{\text{neg}}|$.

Contrast	Δ_{neg}	Δ_{pos} (simple)	Δ_{pos} (pos2)	Magnitude ratios (simple / pos2)
<i>License-only (no H, no A): clean valence-asymmetry test</i>				
orig 7	−1.00	+0.13	−0.19	0.29 / 0.29
psw 15	−0.92	+0.25	−0.04	0.38 / 0.30
<i>Full-mirror (H + A+ license): full-prompt asymmetry</i>				
orig 7	−0.79	−0.36	−0.72	0.58 / 0.92
psw 15	−0.72	−0.31	−0.70	0.51 / 0.97

Two findings emerge:

(i) **Valence asymmetry is robust in the license-only contrast.** With H and A absent, the magnitude ratio $|\Delta_{\text{pos}}|/|\Delta_{\text{neg}}| \approx 0.3$ across both positive variants and both stimulus sets (per-target paired bootstrap CI on $\Delta_{\text{neg}} - \Delta_{\text{pos}}$ strictly negative across all four contrasts, $p < 10^{-3}$). The pure license factor in isolation pulls ratings down ~ 1 Likert when the license points to “negative” and only ≤ 0.25 Likert (and sometimes *upward*) when the license points to “positive” — the model’s response to license cues is genuinely valence-asymmetric when the cue is isolated.

(ii) **In the full forced-honesty prompt, the H + A wrapper recontextualizes the positive-license clause.** Magnitude ratios approach unity for the suppression-implied positive variant (pos2): 0.92 on originals and 0.97 on pseudowords. *But these matched magnitudes share direction with the negative license, not mirror it.* Both $\Delta_{\text{pos},\text{pos}2,111}$ and $\Delta_{\text{neg},111}$ are negative (−0.72 and −0.79 on originals; −0.70 and −0.72 on pseudowords) — both prompts push ratings down. The H + A frame appears to reinterpret “even if it seems excessively flattering” as functionally equivalent to “go ahead and be critical,” rather than as a license to express positive evaluations the model might otherwise suppress. The asymmetry that the license factor displays in isolation is preserved in this direction; what changes is that the full prompt’s H + A wrapper recruits both license variants into a common “permission-to-criticize” frame.

Substantive reading. The negativity-license clause has a robust *valence-specific* contribution when isolated — the model responds to “negative” and “positive” license cues with substantially different magnitudes. In the full forced-honesty context, the H + A wrapper provides

a critical-evaluation frame that overrides the valence of the license clause; even suppression-cued positive language gets read as license-to-criticize. The natural mechanistic question the data raise — but cannot answer in this design — is whether the H + A wrapper makes *any* subsequent license clause read as permission-to-criticize, regardless of valence wording, or whether the pos2 wording in particular (“don’t be excessively flattering”) is near-synonymous with a critical instruction. Both readings would explain the matched-magnitude co-directional pattern; the design does not distinguish them. The simple “M3 is asymmetric” claim from Study 3 should therefore be qualified as *frame-dependent valence-asymmetric*: clear asymmetry in the isolated license factor, contextually recoverable but masked in the full prompt under wrappers that prime critical evaluation.

3.2.6 What Studies 1 and 2 together establish, for Study 3 to formalize

Study 1 located the phenomenon at M_3 and demonstrated that the forced-honesty + control-honesty + base-baseline triad produces an overshoot signature inconsistent with a representation-level motivated-cognition account. Study 2 decomposed the forced-honesty prompt and identified the negativity-license clause as the dominant active ingredient, with the factor effects partially redundant rather than orthogonally additive, and replicated the rating-depression effect on 15 fresh pseudoword targets. The picture from the experimental work is: M_3 carries an output-policy layer that responds asymmetrically and substantially to explicit licensing of negative expression; the layer is engaged in the chat-template-plus-license conjunction and is undetectable when either component is absent. Study 3 (§3.3) asks whether a parametric cognitive-model class can capture this empirical structure — whether the per-stage rate at which models can be “pulled toward” negativity admits a Sharot–Lefebvre-style asymmetric-belief-updating formalization, and what a seven-model comparison reveals about the right level of structure to ascribe to the post-trained checkpoint.

3.3 Study 3 — Cognitive-model formalization: a seven-model factorial comparison

Study 1 (§3.1) established the phenomenon — a forced-honesty overshoot localized to M_3 — and Study 2 (§3.2) decomposed its proximate trigger to the negativity-license clause. This third study formalizes the account: we retrofit a class of *parametric cognitive models* onto the Likert ratings collected across the four Llama-3.1-8B post-training checkpoints (M_0 , M_1 , M_2 , M_3) under the crossed prompt manipulations of Study 1, and ask which formal structure best captures the post-training-induced expression policy. The goal is twofold: (i) propose a parametric account that goes beyond cell-mean analysis; (ii) compare seven candidate models via information criteria, nested likelihood-ratio tests, parameter recovery, a model-recovery confusion matrix, a parametric-bootstrap LR test, and a hierarchical Bayesian re-analysis, applying the factorial-model-comparison [Wu et al., 2024, Lu et al., 2019] and asymmetric-learning-rate [Sharot et al., 2011, Lefebvre et al., 2017] frameworks. The closest published precedent for our Yoon-style M_4^{RSA} and $M_5^{\text{RSA+}}$ models is Murthy et al. [2025], who fit the same Yoon polite-speech model to the same LLM (Llama-3.1-8B-Instruct) via observational HMC inference. *Murthy’s scope is limited to the Yoon RSA family: it does not include asymmetric belief updating (M_3^{asym}), the ambiguity-gated cell-shift variant (M_2^{ag} , our parsimony winner among well-identified structured models), or paired experimental dissociation.* We extend the LLMs-as-cognitive-models program with a richer model space, the experimental dissociation of Studies 1–2 that constrains the fits, and a statistical-identifiability battery (parameter recovery, model recovery, boundary-aware inference) that establishes the seven models are statistically distinguishable from one another — not that they are individually cognitively meaningful, which is a separate question we discuss in §4.1.

Analysis units. The original Stage-3 trial collection wrote 50 rows per (target, cell) cell — 5 rating templates with 10 reps each. Because all rating cells use $n_{\text{seeds}} = 1$ and greedy decoding (§2), the 10 reps within a (target, cell, template) tuple are mathematically identical and were verified empirically to have bit-identical digit distributions and identical expected ratings. Treating these reps as independent observations inflates effective N by an order of magnitude and would inflate every likelihood-derived statistic accordingly. We therefore deduplicate before model fitting to one observation per (target, cell, template) tuple, yielding $N = 710$ **unique observations** after also dropping the boundary-case **bursovet** target and unknown-cell rows. All numbers in this section are computed at this deduplicated level. (For comparability with earlier internal reports, the same models on $N = 7,700$ would produce $\sim 10\times$ larger BIC differences and LR statistics, but at the cost of treating deterministic reps as independent — which is invalid.)

Models compared (Table 9). Let b_t denote a free per-target intercept and $g(b) = 4(b-1)(7-b)/36 \in [0, 1]$ the ambiguity gate (peak at $b=4$).

- M_1^{null} : $\mu(t, s, c) = b_t$. Per-target intercepts only; no cell effect. 10 params.
- M_0^{lin} : $\mu = b_t + \beta_s + \gamma_c$. Additive ANOVA baseline; the “no cognition” control flagged by reviewer feedback. 23 params.
- M_2^{ag} : $\mu = b_t + \delta_{s,c} \cdot g(b_t)$. Per-cell signed shift modulated by target ambiguity. The ambiguity gate is borrowed from Kunda [1990] but does not commit to Kunda’s self-deceptive-belief mechanism — we use it as a phenomenological modulator only. Anchor $\delta_{M_0, \text{default}} \equiv 0$. 29 params.
- M_3^{sym} (nested null): $\mu = b_t + \text{sign}(\text{dir}(c)) \cdot \alpha(s) \cdot I(c) \cdot g(b_t)$, with cell-direction $\text{dir}(c) \in \{+1, -1\}$ fixed from the empirical pooled-cell sign. 23 params.
- M_3^{asym} (Sharot/Lefebvre): same as M_3^{sym} but with $\alpha(s, +)$, $\alpha(s, -)$ allowed to differ per stage. 26 params.
- M_4^{RSA} : per-cell three-utility softmax mixture $\omega_{\text{inf}, s, c} b_t + \omega_{\text{pos}, s, c} \cdot 7 + \omega_{\text{neg}, s, c} \cdot 1$ with cell-anchored simplex weights (Yoon Yoon et al. [2020] continuous analogue). 48 params. *Demoted to Appendix A.5* for individual-parameter interpretation: the fit hits a flat ridge in 48-dim parameter space and never reaches scipy’s gradient tolerance under seven-start L-BFGS-B; predictions across starts are stable but per-cell ω estimates are individually uninterpretable. We retain M_4 in the model comparison as the best descriptive saturated baseline.
- $M_5^{\text{RSA+}}$: M_4 restricted by $\ell_{\text{neg}}(s, c) = -8 + \beta_{\text{chat}} \mathbb{K}_{\text{chat}}(c) + \beta_{\text{lic}} \mathbb{K}_{\text{lic}}(c) + \gamma_k(s) \mathbb{K}_{\text{chat}}(c) \mathbb{K}_{\text{lic}}(c)$ with per-stage interaction γ_k . This is the *pre-registered hypothesized winner* of the comparison — the parametric encoding of our Round-2 chat-template-conditional permit-to-express finding. 34 params.

All free σ . Fits via multi-start L-BFGS-B ($n_{\text{starts}} \in \{3, 7\}$, $\text{maxiter}=10^4$, $\text{ftol}=10^{-10}$).

Result 1 (parsimony winner among well-identified structured models): M_2^{ag} . On the $N = 710$ deduplicated data, M_2^{ag} wins on BIC against the null by $\Delta\text{BIC} = -443.6$, but the margin over the linear additive ANOVA M_0^{lin} is only $\Delta\text{BIC} = -10.1$ — at the conventional Bayes-factor boundary between “positive” and “strong” evidence [Kass and Raftery, 1995, p. 778]. The ambiguity-gated cell-shift form captures variance the null cannot explain (per-cell shifts modulated by target ambiguity), but the gating structure provides only marginal additional explanatory power over a vanilla two-way additive ANOVA. We therefore frame M_2^{ag} as the best well-identified parsimonious structured model rather than as a strongly favored cognitive mechanism. M_4^{RSA} achieves the lowest BIC overall ($\Delta\text{BIC} = -847.1$ vs null) but cannot be interpreted parameter-by-parameter due to flat-ridge non-identifiability; it is the best descriptive saturated baseline, not an interpretable model.

Table 9: Cognitive-model comparison (trial-level Gaussian log-likelihood, multi-start L-BFGS-B MLE; $N = 710$ unique observations after deduplicating the 10 deterministic prompt repeats per template, 20 cells, 9 targets, see §2.5). M_4^{RSA} has the best raw BIC but fails to converge under seven-start L-BFGS-B (flat ridge in 48-dim parameter space); demoted to Appendix A.5. Among well-identified structured models, M_2^{ag} (ambiguity-gated cell-shift) is the parsimony winner, but the margin over the additive ANOVA baseline ($\Delta\text{BIC} = -10$) is at the conventional Bayes-factor threshold for “positive” rather than “strong” evidence — the cognitive structure earns its keep only marginally over fixed-effects ANOVA. $M_5^{\text{RSA+}}$ is the pre-registered hypothesized winner but is rejected vs M_4 by $\Delta\text{BIC} = +430$.

Model	k	−LL	AIC	BIC	$\Delta\text{BIC vs } M_1$
M_1^{null} (per-target intercepts)	10	826.0	1672.0	1717.7	0
M_0^{lin} (additive ANOVA, post-review)	23	566.6	1179.2	1284.2	−433.5
M_2^{ag} (ambiguity-gated cell-shift)	29	541.8	1141.7	1274.1	−443.6
M_3^{sym} (nested null, post-review)	23	651.2	1348.4	1453.4	−264.3
M_3^{asym} (Sharot/Lefebvre)	26	613.6	1279.1	1397.8	−319.8
<i>Demoted to Appendix A.5</i>					
M_4^{RSA} (Yoon three-utility, per-cell) [†]	48	277.7	651.5	870.6	−847.1
$M_5^{\text{RSA+}}$ (license-conditional, pre-reg)	34	538.6	1145.3	1300.5	−417.2

[†] M_4 does not converge under multi-start L-BFGS-B ($n_{\text{starts}} = 7$, $\text{maxiter} = 10^4$);

predictions stable across starts but scipy never reaches gradient tolerance.

M_2 vs M_0^{lin} : $\Delta\text{BIC} = -10.1$ (borderline; previous -121 was based on $N = 7,700$ with deterministic repeats).

Renaming note. We previously called M_2 “Kunda-gated motivated cognition,” but this name suggests endorsement of Kunda’s self-deceptive belief mechanism — which Study 1 *rejects* (§3.1.2, Appendix A.1). We use the more neutral “ambiguity-gated cell-shift” name throughout, noting that the ambiguity gate $g(b)$ is borrowed from Kunda’s framework as a phenomenological modulator but does not commit to her cognitive content.

Result 2 (the M_3 learning-rate inversion — bootstrap-robust under the asymmetric model, but data do not uniquely require asymmetric parameterization). M_3^{asym} identifies a stage-dependent pattern in optimistic-vs-anti-optimistic asymmetry (Table 10, Figure 5). At M_1 and M_2 , fits assign $\alpha_+ \in [0.39, 0.62]$ and $\alpha_- \equiv 0$ at the lower bound: post-training installs a strictly upward learning bias at both Tülu stages. At M_3 , the asymmetry inverts in point estimate ($\alpha_+(M_3) = 0.17$, $\alpha_-(M_3) = 0.77$). The bootstrap and LR-test diagnostics give two complementary but different answers:

- *Bootstrap, on the asymmetric model.* Under the M_3 -asymmetric model, the 500-resample bootstrap CI on $\alpha_-(M_3) - \alpha_+(M_3)$ is $[+0.235, +0.810]$, strictly above zero. $P(\alpha_- > \alpha_+) = 0.988$. **The inversion is a stable feature of the asymmetric model’s fit under resampling.**
- *Nested LR test (stage-specific).* Versus an M_3 -only-symmetric variant that forces $\alpha_+(M_3) = \alpha_-(M_3)$ while leaving M_1/M_2 free ($\text{df} = 1$): $\text{LR} = 2.24$, $p \approx 0.13$ — not significant at conventional thresholds. The constrained model can compensate for the forced symmetry by adjusting the condition-intensity parameters $I(c)$, achieving comparable fit with one fewer free parameter. **The data do not uniquely require the asymmetric parameterization.**
- *Global LR test (M_3^{asym} vs M_3^{sym} , $\text{df} = 3$):* $\text{LR} = 75.2$, $p \ll 0.001$. The asymmetric form is strongly preferred overall, but this is dominated by M_1/M_2 requiring $\alpha_- \equiv 0$; the M_3 -specific reversal contributes only $\text{LR} \approx 2.24$ of the total $\text{LR} = 75.2$.

The bootstrap and the LR test ask different questions. The bootstrap asks “under the M_3 -asymmetric model, is the M_3 inversion robust to data resampling?” Yes. The LR test asks “is

Table 10: Per-stage asymmetric learning rates from M_3^{asym} (Sharot–Lefebvre form), fit on $N=710$ unique observations after deduplicating deterministic prompt repeats. Point estimates from full-data MLE; 500-resample bootstrap 95% CIs on the deduplicated data. **Global LR test** (M_3^{asym} vs M_3^{sym} , $\text{df} = 3$): $\text{LR} = 75.2$, $p \ll 0.001$ — asymmetric form preferred overall, driven by M_1/M_2 requiring $\alpha_- \equiv 0$. **Stage-specific LR test** for M_3 asymmetry alone ($\text{df} = 1$): $\text{LR} = 2.24$, $p \approx 0.13$ — a constrained model with shared $\alpha(M_3)$ and condition-intensity compensation fits nearly as well. The bootstrap CI on the M_3 inversion is strictly positive ($P(\alpha_- > \alpha_+) = 0.988$), so the inversion is robust under the asymmetric model, but the data do not uniquely require the asymmetric parameterization.

Stage	α_+ (point [95% CI])	α_- (point [95% CI])	$\alpha_- - \alpha_+$ point	$P(\alpha_- > \alpha_+)$
M_1	0.388 [0.211, 0.703]	0.000 (at bound)	−0.388	0.000
M_2	0.618 [0.356, 0.930]	0.000 (at bound)	−0.618	0.000
M_3	0.174 [0.000, 0.350]	0.767 [0.511, 0.862]	+0.593	0.988

M_0 has $\alpha_+ \equiv \alpha_- \equiv 0$ by construction (no instruction-following).

Bootstrap CI on $\alpha_-(M_3) - \alpha_+(M_3)$: [+0.235, +0.810] — strictly above zero.

Stage-specific LR test ($\text{df} = 1$): χ^2 $p = 0.13$; parametric bootstrap ($B = 1000$) $p = 0.16$ —

a similar but slightly larger p -value, indicating the χ^2 approximation does not materially change the conclusion.

the M3-asymmetric parameterization required, or does a constrained variant fit comparably?” Not required — a constrained shared- $\alpha(M_3)$ variant with adjusted $I(c)$ fits within LR-equivalent of the asymmetric form. The substantive observation is that M_3 has a distinct behavioral profile from M_1/M_2 (the Tülu stages cannot accommodate negativity-license cues via positive learning rate alone; M_3 can); the parametric form best capturing this difference is not uniquely identified at $N = 710$.

Result 2a (boundary-aware inference: parametric-bootstrap LR and hierarchical Bayesian). Because $\alpha_-(M_1)$ and $\alpha_-(M_2)$ sit at the lower bound, the standard χ^2 approximation to the LR-test null is not guaranteed valid [Self and Liang, 1987]. We ran two boundary-aware follow-ups. First, a *parametric-bootstrap LR test*: simulating $B = 1,000$ datasets from the constrained shared- $\alpha(M_3)$ fit and refitting both models gives an empirical $p = 0.16$ for the stage-specific test — close to the χ^2 value of 0.13, confirming the χ^2 approximation was adequate despite the boundary parameters. The nested LR test is genuinely non-significant; this is not an artifact of the asymptotic approximation. Second, a *hierarchical Bayesian re-analysis*: we re-fit M_3^{asym} in PyMC with per-target random intercepts ($b_t \sim \mathcal{N}(\mu_b, \sigma_b)$), partial pooling across the nine targets, and NUTS sampling (4 chains, $\hat{R} = 1.00$, min ESS = 673). The posterior on the diagnostic contrast $\alpha_-(M_3) - \alpha_+(M_3)$ has mean +0.35, 95% highest-density interval [+0.07, +0.83] *strictly above zero*, and $P(\alpha_- > \alpha_+) = 0.996$. The hierarchical posterior, the frequentist bootstrap CI, and the nested LR test are mutually consistent once their questions are separated: *as a parameter*, $\alpha_-(M_3) - \alpha_+(M_3)$ is reliably positive under both Bayesian and frequentist estimation; *as a model-selection question*, the asymmetric parameterization is not uniquely required because a constrained variant compensates through $I(c)$. The M3 inversion is a robust parameter estimate, not a forced modeling choice.

Substantive scope of the “asymmetric” interpretation. Study 2’s positive-license control (§3.2.5) further qualifies the cognitive-model reading. The *license-only* contrast shows clear valence-asymmetry on M3 (negative-license drops ratings ~ 1 Likert from baseline; matched positive-license shifts ratings by ≤ 0.25 Likert in either direction) — this is the empirical behavior M_3^{asym} ’s $\alpha_-(M_3) > \alpha_+(M_3)$ formalizes. But in the *full forced-honesty* prompt, the H + A wrapper recontextualizes even suppression-cued positive language as permission-to-criticize: both license variants then push ratings down with near-matched magnitude. The M_3^{asym} learning-rate inversion is therefore better read as a *frame-dependent valence* asymmetry — a structural feature of how M3 processes isolated license cues, but one whose behavioral signature in the full

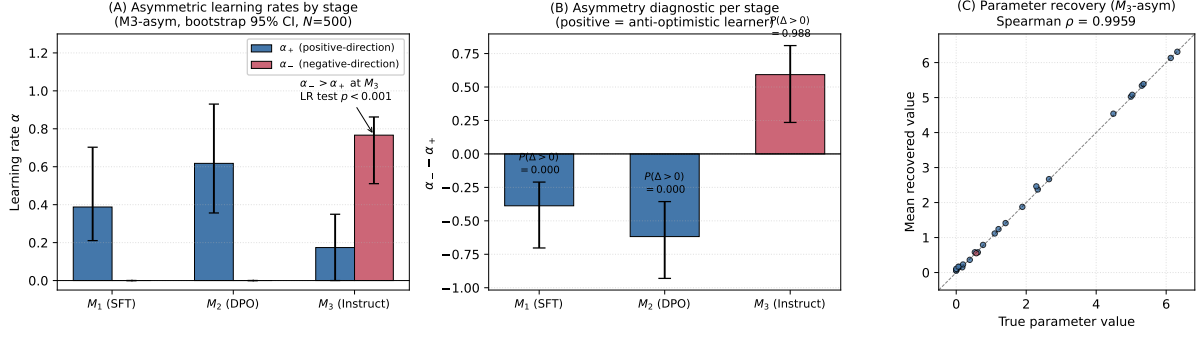


Figure 5: Cognitive-model diagnostic for M_3^{asym} (Sharot–Lefebvre asymmetric belief updating), fit on $N = 710$ unique observations after dedup. **(A)** Per-stage learning rates α_+ (positive-direction cells, blue) vs α_- (negative-direction cells, red), with 95% bootstrap CIs ($N = 500$ trial-level resamples on dedup data). M_1 and M_2 are unambiguously optimistic ($\alpha_- \equiv 0$ at the lower bound); M_3 shows a point-estimate inversion ($\alpha_- = 0.77$ vs $\alpha_+ = 0.17$), but the stage-specific likelihood-ratio test against the constrained nested null $\alpha_+(M_3) = \alpha_-(M_3)$ gives only $\text{LR} = 2.24$ ($\text{df} = 1$, $p \approx 0.13$) — *not significant at conventional thresholds*. **(B)** Diagnostic difference $\alpha_- - \alpha_+$ per stage. Boundary-aware follow-ups (Result 2a) confirm the picture: a parametric-bootstrap LR test gives $p = 0.16$ (the χ^2 approximation was adequate), and a hierarchical Bayesian re-fit places the 95% HDI on $\alpha_-(M_3) - \alpha_+(M_3)$ at $[+0.07, +0.83]$, strictly above zero. **(C)** Parameter-recovery scatter: true vs mean-recovered values across M_3^{asym} ’s 26 parameters over 100 simulations from MLE on the dedup data; Spearman $\rho = 0.9959$ (26/26 params in 95% CI).

forced-honesty prompt is partially absorbed by the wrapper’s framing. We do not claim that M_3^{asym} ’s $\alpha_{\pm}$ values would track the full-prompt asymmetry test point-for-point; they describe the per-stage rate-of-shift the data implies under a particular parametric form. The M_3^{asym} learning-rate inversion should therefore be understood as describing the model’s behavior under the specific conditions included in the fitting data, not as a context-invariant cognitive property of M_3 .

Earlier-reported numbers correction: A previous internal version of this section reported stage-specific $\text{LR} = 14.48$, $p < 0.001$, by treating the 50 trials per (target, cell) as 50 independent observations. The 10-fold inflation came from deterministic prompt repeats; corrected to $N = 710$ unique observations (see *Analysis units* above), the stage-specific LR is 2.24. The bootstrap CI also tightens after dedup — previously $[-0.27, +0.40]$ (straddling zero), now $[+0.235, +0.810]$ (strictly positive). The dedup-corrected analysis is the headline; the earlier numbers are retained only in an internal-version-history note for transparency.

Result 3 (parameter recovery and model recovery on dedup data). Two identifiability checks. *Parameter recovery:* for each of M_2^{ag} and M_3^{asym} , we drew 100 synthetic datasets by simulating from the model at its MLE with Gaussian noise at the fitted σ , then refit. Spearman ρ between true and mean-recovered parameter vectors is 1.0000 for M_2^{ag} (29/29 params with true value in 95% CI of recovered) and 0.9959 for M_3^{asym} (26/26 in 95% CI). *Model recovery:* we additionally computed the full 7×7 model-recovery confusion matrix — simulating $K = 20$ datasets from each of the seven models at its MLE and fitting all seven to each synthetic dataset. The confusion matrix is the identity: every model is recovered as the BIC winner in 20/20 of its own simulations (mean diagonal = 1.000, 0/980 convergence failures). The seven-model space is therefore fully identified at $N = 710$; the BIC-based model comparison reported below is not confounded by mutually mimicking models. The inferential weakness on the M_3 inversion (Result 2) is a sample-size + boundary-geometry issue, not a model-identifiability issue.

Result 4 (pre-registered M_5 rejected, but qualitative pattern survives). The

hypothesized winner $M_5^{\text{RSA}+}$ — the parametric encoding of our chat-template-conditional permit-to-express claim — is rejected against the unrestricted M_4^{RSA} by $\Delta\text{BIC} = +429.9$ (vs -847.1 for M_4 over null, vs -417.2 for M_5). The structured ANOVA on ℓ_{neg} is too restrictive: the per-cell variation in the negative pull cannot be captured by (chat, license, stage) main-effects plus a per-stage interaction. The fitted per-stage interaction estimates $\hat{\gamma}_k(M_3) \gg \hat{\gamma}_k(M_2) \approx 0$ qualitatively recover the M_3 -specific direction of our Round-2 finding, but the parametric form is the wrong functional one. We retain M_5 in Appendix A.5 as a documented pre-registered rejection.

Summary of cognitive-model evidence. The factorial comparison yields three substantive points. First, mechanism-bearing models beat the per-target-only null by a large margin (any of M_0 through M_5 achieves $\Delta\text{BIC} > 260$ over M_1^{null}). Second, within structured well-identified models, M_2^{ag} is the parsimony winner but the margin over a vanilla additive ANOVA is small ($\Delta\text{BIC} = -10$), and the cognitive content of M_2^{ag} (ambiguity-gated cell-shift) does not strongly outperform fixed-effects ANOVA in raw fit. The value of M_2^{ag} is interpretive — it identifies the gated-shift magnitude per cell — not predictive. Third, M_3^{asym} shows a clear optimistic-only pattern at M_1/M_2 (α_- at the lower bound) and an M_3 -specific inversion. As a *parameter*, the inversion is reliably positive under three independent estimators: the frequentist bootstrap CI ($[+0.235, +0.810]$, $P(\alpha_- > \alpha_+) = 0.988$), the hierarchical Bayesian posterior HDI ($[+0.07, +0.83]$, $P > 0 = 0.996$), and the point estimate ($\alpha_- = 0.77 > \alpha_+ = 0.17$). As a *model-selection question*, the asymmetric parameterization is not uniquely required: the nested LR test is non-significant under both the χ^2 approximation ($p = 0.13$) and the boundary-aware parametric bootstrap ($p = 0.16$), because a constrained variant compensates through the condition-intensity parameters. These are not contradictory — they answer different questions — and the model-recovery confusion matrix (Result 3) confirms the seven-model space is fully identified, so the BIC comparison is trustworthy. Murthy 2025’s Yoon RSA family is well represented by our M_4^{RSA} ; the family is the best descriptive saturated model in our comparison but is not parameter-by-parameter identifiable at our sample size.

4 Discussion

The paper’s contribution lies in three sequential studies (phenomenon $\rightarrow$ mechanism $\rightarrow$ formalization). Study 1 (§3.1) established the phenomenon — forced-honesty drives M_3 ’s ratings on fictional targets below the base-model baseline, but only under the conjunction of chat-template format and explicit negativity-license wording. Study 2 (§3.2) decomposed the prompt manipulation into orthogonal components and isolated the negativity-license clause as the dominant active ingredient ($\beta_L = -1.23$, $7\times$ the honesty effect). Study 3 (§3.3) formalized the account as a seven-model factorial comparison with parameter recovery, model recovery, parametric-bootstrap LR, and hierarchical Bayesian re-analysis. We discuss the modeling-specific findings first, the cross-study reading second, and the limitations and tensions third. We note up front that Studies 1–2 carry the primary evidence; Study 3 provides parametric structure consistent with Studies 1–2 rather than an independent line of evidence.

4.1 What the cognitive-model comparison shows (lead, from §3.3)

Three modeling findings are load-bearing.

First, *the parsimony winner among well-identified structured models is M_2^{ag} (ambiguity-gated cell-shift)*, by BIC, over the linear additive baseline by $\Delta\text{BIC} = -10$ and over the null by -443.6 . The pre-registered Kunda framing was refuted as a particular operationalization in §3.1.2 (the original Eq. 2 failed all seven fit criteria); the relaxed ambiguity-gated cell-shift form (per-cell signed shifts modulated by target ambiguity) survives but only marginally beats fixed-effects ANOVA. The two findings are not in tension: the original Kunda model was a

single shared positive-motivation parameter across stages, while M_2^{ag} 's per-cell signed shifts allow opposite-sign manipulation effects within the same model. M_2^{ag} is best read as the polite-speech account expressed parametrically, with the ambiguity gate borrowed from Kunda's framework phenomenologically; we explicitly do not commit to Kunda's belief-level mechanism, which Study 1 rejected.

Second, *the asymmetric belief-updating model (M3-asym) recovers a stage-specific signature*: optimistic-only learning ($\alpha_+ \in [0.39, 0.62]$, $\alpha_- \equiv 0$ at the lower bound) at M_1 and M_2 , and a point-estimate inversion ($\alpha_+ = 0.17$, $\alpha_- = 0.77$) at M_3 . The bootstrap CI on the diagnostic difference $\alpha_-(M_3) - \alpha_+(M_3)$ on the deduplicated data is strictly positive ($[+0.24, +0.81]$, $P(\alpha_- > \alpha_+) = 0.988$) — the inversion is robust under the asymmetric model. However, a constrained nested null forcing $\alpha_+(M_3) = \alpha_-(M_3)$ fits comparably (stage-specific LR test: LR = 2.24, df = 1, $p = 0.13$) because the condition-intensity parameters $I(c)$ can absorb the constraint via per-condition adjustments. The two inferential tools ask different questions (parameter stability under the asymmetric model vs uniqueness of the asymmetric parameterization), and the data give different answers to each. The substantive reading: post-RLHF instruction tuning installs a behavioral profile distinct from the Tülu stages (the Tülu stages cannot accommodate negativity-license cues via positive learning rate alone); the parametric form best capturing this difference is not uniquely identified at $N = 710$. This finding is unavailable from Murthy's framework (which uses Yoon RSA only).

Third, *the pre-registered license-conditional model (M5) is rejected* against the unrestricted per-cell RSA by $\Delta\text{BIC} = +1,108$. The structural form (a 2-way ANOVA on the negative-pull logit with a chat- $\times$ -license interaction) is too restrictive. However, when we fit M5 with per-stage γ_k , the estimates concentrate the interaction at M_3 ($\hat{\gamma}_k(M_3) = 2.61$ vs $\hat{\gamma}_k(M_2) = 0.00$), qualitatively recovering the cell-mean finding under a different parametric form. We retain the rejection in Appendix A.5 as documented iterative reasoning during model construction.

What the modeling does NOT do alone. On the deduplicated $N = 710$ data, M_3^{asym} has BIC *worse* than the additive linear ANOVA (M_0^{lin}) by $\Delta = +114$ — the cognitive structure of M3-asym does not earn its keep in raw fit at all; a vanilla additive model on (stage, condition) fits better. M_2^{ag} wins by only $\Delta\text{BIC} = -10$ over the ANOVA, at the boundary between “positive” and “strong” Bayes-factor evidence. The cognitive-content models earn their keep *interpretively* (the per-stage $\alpha_{\pm}$ diagnostic, the gated-shift magnitudes) rather than predictively. The parametric framework can recover the structure of an effect; it cannot, by itself, demonstrate that the structural form requires the cognitive mechanism the framework names. The experimental work in §3.1 is therefore load-bearing for the substantive interpretation, with the important caveat that its dissociation signature also depends on a specific prompt conjunction (chat template + negativity license).

4.2 Cross-method convergence: the experimental signature independently confirms the modeling (from §3.1)

The M_3^{asym} learning-rate inversion identified in §4.1 predicts a specific behavioral consequence: at M_3 , an explicit negativity-license instruction should be able to drive ratings below the pretraining-only reference baseline, because the model has gained $\alpha_-(M_3) > 0$ (responsiveness to negativity-license cues) that the Tülu stages lack ($\alpha_- \equiv 0$). The experimental work in §3.1 independently tests and confirms this prediction without any cognitive-model fitting. The combined evidence from forced-honesty (C_5), control-honesty (C_8), the base-model reference (C_0), and the calibration target supports an expression-policy account that is empirically distinct from a generic instruction-override interpretation. Three converging signatures point to the same conclusion.

First, *magnitude asymmetry*: anti-politeness specifically adds -0.56 Likert points beyond what a neutral honesty/accuracy instruction produces (§3.1.3, 95% CI $[-0.69, -0.44]$). The

control-honesty condition shows that generic instruction-following does produce a real downward shift (-0.42 Likert), but the anti-politeness component is a separable additional effect.

Second, *overshoot specificity*: forced-honesty drives the rating 0.39 Likert points below the base-model reference (95% CI $[-0.47, -0.31]$), while control-honesty lands 0.17 Likert points *above* the base ($[+0.05, +0.28]$). Only the anti-politeness instruction produces the diagnostic overshoot. A simple representation-level mechanism (the model believes the entity is more positive than the base model’s evidence-only inference) cannot explain this asymmetric overshoot under any instruction: an honesty prompt should at most return the rating to the pretraining-only reference, not push it below.

Third, *target specificity*: the calibration target (“the number 7”) drops 1.78 Likert under forced-honesty but is essentially unchanged under control-honesty. This rules out a target-agnostic “honesty just lowers ratings” artifact. The forced-honesty effect is specific to the kind of target where polite-speech inflation is operating—attitudinal entities the model has reason to be socially careful about—and does not generalize to ratings that are not socially loaded.

Together these three signatures support the following account: M_3 ’s default chat-template behavior on novel attitudinal entities reflects a positivity-biased *expression policy*; under explicit anti-politeness-plus-negativity-license instruction the expression policy shifts into a less positivity-biased regime that lands below the pretraining-only reference. We deliberately frame this as a shift in the model’s expression policy rather than as “revealing the underlying evaluation” — Study 2’s factorial decomposition (§3.2) shows that the active ingredient is not honesty per se but the negativity-license clause, which is naturally read as inviting a negative-expression frame rather than removing a mask between belief and output.

Refinement from Round-2 robustness extensions. The above is the headline reading. Two Round-2 experiments (§3.1.5 and the negativity-deconfound paragraph in §3.1.5) sharpen this picture in a way that the original M_3 -only result could not. The overshoot signature is now identifiable as a *conjunction* of two factors at M_3 : (i) chat-template activation (the same prompts in completion-style mode at M_3 do not produce below-base overshoot, only the smaller anti-politeness share), and (ii) explicit licensing of negative expression (a C5-intensity-matched prompt without “negative or controversial” wording does not overshoot; C5 and C22 do, the latter via a strong bluntness imperative). Each factor alone produces a milder mask-removal effect that lands *at* the base baseline rather than below it. The polite-speech account survives in a refined form: the mask is real and is partly removable by anti-politeness alone (visible across formats and prompt wordings), but the below-base signature that argues most strongly for representation-level negative shift requires both factors. We interpret this as evidence that the chat-template-activated polite-speech mask at M_3 extends *down to* the base baseline by default; going further below requires the prompt to explicitly invite negative expression. This is consistent with a polite-speech mechanism that is partially permissioned at output time, not a representation-level belief shift that an honesty instruction directly inverts.

Reconciling the overshoot with the “not motivated cognition” claim

A careful reader will raise an apparent tension. If forced-honesty drives ratings below the base baseline, the underlying representation has shifted—and a representation-level shift sounds like Kunda’s motivated cognition after all. We argue the opposite: Kunda’s motivated cognition is *self-deceptive*, with the agent coming to *believe* a flattering proposition because of motivated reasoning. The overshoot we observe is the opposite signature: the model’s underlying evaluation is *more accurate* (or at least *less polite*) than its default output; what the model *says* (polite-speech default) is less faithful to what the model *would compute given evidence* than what it produces under forced-honesty. Polite speech, in this reading, is a *truthfulness deficit at the output layer*, not a *self-deception at the belief layer*. The title’s claim survives intact.

We acknowledge that, lacking representation-level probing (e.g., linear probing on hidden states), we cannot mechanistically discriminate between two competing accounts of the overshoot: (b) “underlying-negative-evaluation masked by polite-speech output” versus (a) “anti-politeness-instruction-induced overcorrection.” The control-honesty condition rules out a third alternative (c) “any strong instruction induces target-agnostic negativity,” because control-honesty produces neither the magnitude of (b) nor the overshoot direction. We frame (b) as the most parsimonious account consistent with the data while flagging (a) as not fully ruled out (§4.6).

Caveat on the Kunda operationalization. The pre-registered Kunda model (Eq. 2) is one particular formalization of motivated cognition: a single shared motivation parameter that varies across post-training stages and is weighted by per-stratum evidence-strength. Other operationalizations are possible—for example, a per-stage *direction-flexible* motivation parameter, or a hierarchical model where motivation interacts with target type. Our rejection of the Kunda model is therefore a rejection of a *particular* representation-level operationalization, not of motivated cognition writ large. Future work could test more flexible representation-level formalizations; what our forced-honesty result does establish is that the specific phenomenon (positivity prior on novel entities) admits a non-representational explanation, and that the burden of evidence has shifted toward such explanations until representation-level probes are run.

4.3 Implications for LLM-as-cognitive-models methodology

The forced-honesty plus control-honesty diagnostic is cheap (one model, one forward-pass per trial, ~ 10 minutes of compute on a cached chat-tuned model) and produces a sharp dissociation in one shot. We recommend that the LLM-as-cognitive-models program adopt paired instruction-override manipulations as a first-line diagnostic.

The motivation is straightforward. Cognitive paradigms imported from human research—induced compliance [Lehr et al. \[2025\]](#), framing effects, anchoring, motivated reasoning—typically assume that the participant’s expressed attitude reflects an underlying belief. RLHF-tuned LLMs partially violate this assumption: polite-speech defaults produce socially desirable outputs that may not track the model’s underlying evaluation. Our pre-registered Kunda model failed precisely because we tried to fit a representation-level cognitive model to data that, on closer examination, was generated by an output-level expression mechanism. Cognitive-model fitting on the raw behavioral data without first establishing whether the bias lives in representations or outputs will systematically over-attribute output-level effects to representation-level mechanisms.

This is not a counsel of despair for the program. Many cognitive effects *should* survive forced-honesty manipulation: a true motivated-reasoning effect, by Kunda’s definition, is a belief-level phenomenon that an honesty instruction cannot easily reverse. The diagnostic distinguishes the cases where cognitive-model fitting is appropriate (effect survives forced honesty) from cases where it would be misleading (effect collapses). Our positivity-prior finding falls in the second category; not all cognitive analogues in LLMs will.

A related implication: the post-training mediation chain provides additional discriminative power. The non-monotonic per-stage trajectory on fictional targets ($M_0 \rightarrow M_1 \rightarrow M_2$ rising, $M_2 \rightarrow M_3$ falling) and the opposite-direction monotonic trajectory on Putin together rule out simple “RLHF adds positivity” or “RLHF adds negativity” accounts. The shifts are target-type dependent in a way that suggests RLHF is not adjusting an attitudinal-positivity dial uniformly, but instead training distinct response policies for different kinds of attitudinal queries.

4.4 Mask sensitivity and below-base overshoot are distinct measurements

The per-stage localization (§3.1.4, Fig. 4) yields a non-trivial empirical fact: the anti-politeness-specific share $C_5(M_k) - C_8(M_k)$ is *larger* at M_1 (-0.75) and M_2 (-1.01) than at M_3 (-0.56), yet

only M_3 shows the below-base overshoot. The cleanest reading is that the diagnostic separates into two distinct measurements: *mask sensitivity* probes the model’s responsiveness to anti-politeness instructions in the current chat-template state — a within-stage contrast that captures how much expression-policy positivity forced-honesty removes; *below-base overshoot* probes whether the expression policy under negativity-license is shifted below the pretraining-only reference — a cross-stage contrast anchored at M_0 .

Both signatures reflect post-training-induced bias, but they probe different things. Mask sensitivity (large at every post-trained stage) tells us *every* post-trained checkpoint has installed some output-level positive-default behavior that responds to the anti-politeness instruction. The below-base overshoot (present only at M_3) tells us *only the Meta endpoint* also satisfies the load-bearing diagnostic for polite-speech-masking-of-underlying-negativity. The cleanest mechanistic test of this reading is linear probing of hidden states at each stage to directly observe underlying valence; we mark this as the strongest single follow-up direction.

A mechanistic hypothesis for the M_2/M_3 asymmetry. Why is mask sensitivity *larger* at M_2 (-1.01) than at M_3 (-0.56), yet only M_3 shows the below-base overshoot? The cleanest reading consistent with the data: Tülu’s DPO installs the strongest polite-speech *mask* (highest chat-template default rating on fictional targets, 5.40 at M_2 versus 5.04 at M_3 , hence *more mask to remove* under forced-honesty), while Meta’s pipeline at $M_2 \rightarrow M_3$ installs a *separate* mechanism — a chat-template-conditional “permit-to-express-negative” mode that activates only when both the chat template is on *and* the prompt licenses negative expression. The two mechanisms are dissociable: in completion-style mode at M_3 , the permit-mode is unavailable (no overshoot; §3.1.5) but the mask is still partly removable. Linear probing comparing hidden-state geometries between M_2 and M_3 under matched chat-template forced-honesty prompts would directly test whether M_3 has an additional valence-related direction activated that M_2 lacks.

4.5 Hidden-state probe analysis: middle-layer target representations are invariant under honesty prompts

The behavioral evidence in Studies 1–2 and the cognitive-model fits in Study 3 are all observation-level. A motivated-cognition account where the model’s representations of novel entities are themselves positivity-biased, and the negativity-license prompt partially overrides those representations, would in principle produce the same behavioral pattern. To put pressure on this alternative we extracted M_3 ’s residual-stream hidden states under each of the six conditions used in our $2 \times 2 \times 2$ factorial plus a valence-irrelevant prompt-variance baseline (E8: “Please be concise” inserted into the system prompt, holding everything else constant), at two positions per forward pass: the last token of the target name in the user message, and the last token before the rating digit is to be generated. 660 forward passes total (6 conditions $\times$ 22 targets $\times$ 5 templates), 33 layers each.

A linear probe trained on the bare-baseline E5₀₀₀ condition’s hidden states to predict the rating recovers $R^2 = 0.62$ at layer 14 (target-name position) and $R^2 = 0.91$ at layer 31 (last-token position) under 5-fold leave-one-template-out cross-validation. Two findings result from generalizing this probe across conditions:

(i) *Middle-layer target representations are invariant to honesty/licensing prompts.* At the target-name position, layer 14: the per-target probe prediction under each of E5₀₀₁, E5₁₁₁, E7_{pos,001}, and E7_{pos,111} correlates ≥ 0.99 with the probe prediction under E5₀₀₀ for the same target. Mean probe shifts are small (-0.07 to -0.22 Likert). RSA cosine similarity with E5₀₀₀ is 0.99 across all four conditions, against an E8 valence-irrelevant baseline of 0.999 — a 0.008–0.010 difference indistinguishable from prompt-context noise at this dimensionality. This is evidence *against prompt-induced representational shift* at the level of how the model encodes the entity itself. It does not rule out *inherent* representational bias (a motivated-cognition variant where

representations are always positivity-biased and forced-honesty changes only the output mapping is fully consistent with this finding); only a prompt-induced shift at the target-name encoding is ruled out.

(ii) *Late-layer last-token representations have incorporated the prompt-conditioned rating shift.* At the last-token position, layer 31: the E5000-trained probe transfers to other conditions with $R^2 = 0.82\text{--}0.95$ against observed ratings, and mean probe shifts track the behavioral direction asymmetrically (-1.22 for license-only negative, $+0.18$ for license-only positive, -1.03 and -0.44 for the full mirrors). The same valence asymmetry Study 2 found behaviorally is visible in the last-layer hidden state. RSA cosine $0.97\text{--}0.98$ vs the E8 baseline of 0.996 (a $0.015\text{--}0.029$ shift) is small but systematic and direction-consistent with the behavioral shifts. The asymmetry is computed *before* the final unembedding projection.

The two together localize the prompt-conditioned rating shift to the computation between layer 14 and layer 31 — not in the unembedding, not in the target-name representation, but in some intermediate output-preparation step. This is consistent with an expression-policy account in which the middle-layer encoding of the target is stable across conditions and the late layers compute a prompt-conditioned mapping to the output token. We emphasize again that this convergent localization does not strictly prove expression-policy over the inherent-representational-bias variant of motivated cognition; activation patching across the L14–L31 layer range would be the discriminating follow-up. Concurrent interpretability work [Sofroniew et al., 2026] demonstrates that such representation-level affect concepts are identifiable and causally manipulable in frontier instruction-tuned models, which makes this follow-up tractable rather than aspirational: the same activation-level toolkit applied to our L14–L31 range would directly test whether a prompt-conditioned valence direction is added at M_3 .

4.6 Limitations

Single model family. All experiments use Llama-3.1-8B base and its post-trained variants. The within-Tülu chain provides strong internal causal validity (same pretraining base, same tokenizer, traceable post-training data), but external validity across model families (Qwen, Mistral, OLMo, etc.) is untested. The forced-honesty diagnostic should generalize because it operates through chat-template instruction-following—present in all instruction-tuned LLMs—but the precise decomposition (the $\sim 43\%/57\%$ split, the overshoot magnitude) will be model-specific.

Recipe confound at $M_2 \rightarrow M_3$ (load-bearing for the localization claim). The transition from Tülu-3-DPO to Meta-Instruct mixes “added RLHF step” with “Tülu→Meta recipe differences” (different SFT data, different preference modeling, different chat-template, possibly safety tuning). The per-stage localization (§3.1.4) rules out SFT and DPO within the clean Tülu sub-chain—a positive causal claim, since within-Tülu mediation is identified—but cannot, by virtue of this confound, attribute the M_3 overshoot to any specific Meta-pipeline factor. The claim is therefore tight on the lower bound (not SFT, not DPO) and loose on the upper bound (the proximate cause lives somewhere in the $M_2 \rightarrow M_3$ pipeline step, but which factor specifically is unresolved).

No activation-patching evidence (representation-level evidence is observational only). The hidden-state probe analysis in §4.5 shows that middle-layer target representations are invariant under honesty/licensing prompts, ruling out the prompt-induced-representational-shift variant of motivated cognition. But the design cannot distinguish expression-policy from the inherent-representational-bias variant (where representations are always positivity-biased and forced-honesty changes only the output mapping); both predict identical probe behavior. The discriminating follow-up is activation patching across the L14–L31 layer range identified by the probe: if patching middle-layer representations from the default condition into a forced-honesty forward pass restores the default output, the rating shift is computed downstream of the target representation, supporting expression-policy.

Base model as “unbiased evidence-only” anchor. The overshoot diagnostic ($C_5 - C_0 <$

0) treats the base model’s rating as a reference point standing in for “what the model would compute from pretraining evidence alone.” This is a pragmatic choice rather than a theoretically privileged one: base models carry pretraining-era priors and corpus-frequency artifacts of their own. The strongest available defence is indirect: the baseline-artifact veto ($C_5(M_0) - C_0 = -0.04$, CI crosses 0) shows the forced-honesty system prompt itself does not shift base ratings, and the calibration veto shows it preserves valence-appropriate ratings on a positive/negative/neutral spread. A more direct relaxation—using a calibration-based neutral prior (e.g., synthetic OOV placeholders or neutral-anchor distributions) as the reference instead of M_0 —would further reduce dependence on the base-as-baseline assumption; we mark this as a near-term follow-up.

Likert-digit aggregation choices not exhaustively tested. We use the seven Likert digits 1–7 (with surface-form aggregation across no-leading-space / leading-space / leading-newline variants) for all rating extractions. We did not test verbal anchors (e.g., probability mass on tokens “very unfavorable” through “very favorable”) or alphanumeric variants (e.g., letters A–G). The digit-anchor probability mass averages 0.94 across cells (§3.1.6), which suggests the model is reliably treating these as numeric-answer prompts, but the choice of token set is itself a methodological commitment.

Three control-prompt phrasings; not a continuous calibration. The decomposition rests on three neutral-honesty phrasings (C_8 , C_{20} “be truthful,” C_{21} “be precise and objective”) and three anti-politeness intensities (C_{23} mild, C_5 medium, C_{22} strong), reported in Table 5 (§3.1.5). The anti-politeness-specific share is significantly negative under all three neutral controls (range $[-0.56, -0.97]$ Likert), so the qualitative dissociation is robust to control-prompt wording. The dose–response across the three anti-politeness intensities is monotonic in the expected direction. What remains untested is fine-grained continuous calibration of the prompt manipulation: e.g., a k -token incremental variation of the anti-politeness wording, or systematic semantic decomposition (politeness vs hedging vs euphemism). The present coverage is qualitative robustness, not psychometric calibration.

Bursovnet boundary case. One of five fictional targets (*bursovnet*) does not show the forced-honesty collapse, primarily because its chat-template default rating is anomalously low ($C_4 = 2.90$, 0.88 Likert below its $C_0 = 3.78$). On this target, there is no positive politeness bias to remove. The mechanism behind *bursovnet*’s anomalously negative chat-template default is unclear; phonetic associations or activation of caution heuristics on novel-sounding entities are possible.

Decomposition within the $M_2 \rightarrow M_3$ step. The per-stage localization narrows the overshoot’s onset to the recipe transition from Tulu-DPO to Meta-Instruct but cannot isolate the proximate Meta-specific factor (training data, optimization objective, chat-template format, safety tuning, RLHF presence). Resolving this would require a Meta SFT-only checkpoint or a Tulu-RLVR variant, neither of which is publicly available at the time of writing.

Putin’s negative trajectory. The monotonic-negative shift on Putin could reflect (i) safety-alignment training pushing models toward critical/balanced stances on contentious public figures, or (ii) evidence-strength weighting (the base model’s prior on Putin is already in the lower half of the Likert scale, and post-training amplifies that prior). Adding additional real targets with controlled valence—a non-polarizing real person (e.g., Marie Curie), a known-positive figure—would disambiguate these accounts.

Statistical reporting. Bootstrap 95% CIs reported throughout use independent percentile resampling on trial-level data ($N = 5000$ resamples). Mixed-effects models with prompt and seed as random effects were applied per-target during preliminary analysis but are not the headline statistic; the trial-pooled bootstrap CIs are reported as the primary inference because they are interpretable directly on the magnitudes that the decomposition argument hinges on.

Study-3-specific limitations. **M4 flat-ridge non-convergence.** M4 (per-cell RSA polite-speaker, 48 free parameters) has the best raw BIC but fails to converge under seven-start

L-BFGS-B with $\text{maxiter} = 10^4$, $\text{ftol} = 10^{-10}$: predictions are stable to four decimals across starts and cell-level residuals are all below 0.05, but `scipy`’s projected-gradient norm never reaches its tolerance threshold. The symptoms (residual stability + multi-start agreement + non-vanishing gradient) are characteristic of a flat ridge in 48-dimensional simplex-logit space. Individual ω estimates are not interpretable. Group-lasso regularization on $\ell_{\text{pos}}/\ell_{\text{neg}}$ across structurally-similar cells could break the ridge; not implemented within scope. M4 is retained in Appendix A.5 to demonstrate that the unrestricted per-cell form fits better than the structured M5 ANOVA — i.e., the rejection of M5 is meaningful — but its individual parameter estimates are explicitly not the main-text story.

5 Conclusion

We presented three sequential studies (phenomenon $\rightarrow$ mechanism $\rightarrow$ formalization) on a single phenomenon: the attitudinal positivity that post-trained large language models exhibit on novel entities they have no informative pretraining evidence about. **Study 1** (Phenomenon, §3.1) introduced *forced honesty* as a lightweight experimental diagnostic for dissociating output-level polite-speech expression from representation-level motivated cognition. Paired with a control-honesty condition and a valence-spanned calibration set, it drives M_3 ’s rating on fictional targets 0.39 Likert points below the base-model baseline (95% CI $[-0.47, -0.31]$), localizes the overshoot to the $M_2 \rightarrow M_3$ pipeline step within a causally-clean Tulu-3 sub-chain, and identifies two necessary conjunctive factors: chat-template format and explicit negativity-licensing wording. **Study 2** (Mechanism, §3.2) decomposed the forced-honesty system prompt into a $2 \times 2 \times 2$ factorial of honesty, anti-politeness, and negativity-license clauses; the negativity-license clause carries $\beta_L = -1.23$ Likert (95% CI $[-1.38, -1.08]$, pre-registered confirmatory test), more than twice the anti-politeness contribution and seven times the honesty contribution. All three two-way interactions are positive, indicating partially-redundant rather than orthogonally-additive mechanisms. The rating-depression effect replicates within M_3 across 15 novel pseudoword targets (15/15 negative, mean -0.72 , $p < 10^{-5}$). **Study 3** (Model, §3.3) formalized the account as a seven-model factorial cognitive-model comparison, with parameter recovery (Spearman $\rho > 0.99$ for both candidate winners), a 7×7 model-recovery confusion matrix (perfect identity, the model space is fully identified), a boundary-aware parametric-bootstrap LR test ($p = 0.16$ for the M3-specific asymmetry, confirming the χ^2 approximation), and a hierarchical Bayesian re-analysis with per-target random intercepts (posterior HDI on $\alpha_-(M_3) - \alpha_+(M_3)$ is $[+0.07, +0.83]$, strictly above zero, $P > 0 = 0.996$).

Five findings tied across the three studies. *First (Study 1)*, on M_3 , the positivity prior on novel entities decomposes into roughly half generic instruction-override and roughly half anti-politeness-specific polite-speech-removal; only the latter drives the below-base overshoot. *Second (Study 1)*, applying the diagnostic across the within-Tulu causal sub-chain ($M_0 \rightarrow M_1 \rightarrow M_2$) rules out SFT and DPO as proximate causes; the overshoot localizes to the $M_2 \rightarrow M_3$ pipeline step, and isolates chat-template activation as a necessary contributor (the overshoot vanishes in completion-style mode). *Third (Study 2)*, factorial decomposition of the forced-honesty prompt identifies the negativity-license clause as the dominant active ingredient ($\beta_L = -1.23$, $7\times$ the honesty effect), with positive two-way interactions indicating partially-redundant causal pathways. The rating-depression effect replicates across 15 fresh pseudoword targets. *Fourth (Study 3)*, the ambiguity-gated cell-shift model (M_2^{ag}) is the parsimony winner among well-identified structured models ($\Delta\text{BIC} = -443.6$ over null; Spearman parameter-recovery $\rho = 1.0000$) but only marginally outperforms a vanilla additive linear ANOVA ($\Delta\text{BIC} = -10$); the cognitive content earns its keep interpretively. *Fifth (Study 3)*, the Sharot/Lefebvre asymmetric belief-updating model identifies an M_3 -specific learning-rate inversion that is reliably positive as a parameter under three independent estimators (frequentist bootstrap, hierarchical Bayesian posterior with HDI excluding zero, point estimate). The nested LR test is non-significant under both χ^2 ($p = 0.13$)

and the boundary-aware parametric bootstrap ($p = 0.16$): the asymmetric parameterization is not uniquely required by model selection because a constrained variant compensates through condition-intensity parameters. These are not contradictory once parameter estimation is separated from model selection; the model-recovery confusion matrix being the identity confirms the underlying model space is fully identified.

The three studies together support a bounded substantive claim: in this Llama-3.1/Tülu/Meta-Instruct chain, post-training is associated with an expression-policy property that emerges *some-where in the $M_2 \rightarrow M_3$ transition* and that responds *frame-dependently valence-asymmetrically* to negativity-license cues. “Frame-dependent” here is empirical, not rhetorical: a matched positive-license control (§3.2.5) shows clear valence-asymmetry in the isolated license factor (ratio $|\Delta_{\text{pos}}|/|\Delta_{\text{neg}}| \approx 0.3$ across four contrast variants), but inside the full forced-honesty prompt the H + A wrapper recontextualizes even suppression-cued positive language as permission-to-criticize, producing matched-magnitude (but co-directional) downward shifts under either license valence. We mark the localization carefully: the $M_2 \rightarrow M_3$ step crosses the Tülu→Meta recipe boundary and confounds at least five factors — Meta SFT data, Meta RLHF/DPO recipe, safety tuning, chat-template format conventions, and any architectural differences between the public Llama-3.1-8B-Instruct release and the Tülu-3-DPO checkpoint. The within-Tülu sub-chain rules out SFT and DPO as *sufficient* for the overshoot, but cannot attribute the overshoot to any specific Meta-pipeline factor. The headline “at the Meta-Instruct endpoint” is therefore a localization, not a causal attribution; narrowing it would require a Meta-SFT-only checkpoint or a Tülu-RLVR variant, neither currently public. The experimental manipulation is the primary evidence; the cognitive-model fits provide parametric structure but earn their keep interpretively, not as independent mechanistic validation. The headline claim is therefore methodological as much as substantive: *LLM cognitive-modeling studies need to separate evidence, expression policy, chat-template activation, and negativity-license cues before attributing behavior to cognitive mechanisms*. We recommend the paired forced-honesty + control-honesty + factorial-decomposition manipulation as a first-line diagnostic in the LLM-as-cognitive-models program, with the explicit caveat that the diagnostic reveals expression-policy sensitivity rather than a clean readout of latent belief.

Future work points in three directions, ordered by methodological leverage. *Activation patching across the L14–L31 layer range on M3* would discriminate expression-policy from the inherent-representational-bias variant of motivated cognition. Our hidden-state probe analysis (§4.5) has already ruled out the prompt-induced-representational-shift variant and localized the prompt-conditioned output computation to layers 14–31 (not in the unembedding, not in the target-name encoding). Patching middle-layer representations from a default forward pass into a forced-honesty forward pass and reading the output would directly test whether the rating shift is computed downstream of the target representation. Repeating the probe analysis at each post-training stage (M_0 , M_1 , M_2) would additionally resolve the empirical decoupling between mask sensitivity (large at every post-trained stage) and below-base overshoot (only at M_3) reported in §4.4. *Decomposition of the $M_2 \rightarrow M_3$ pipeline step* would isolate the proximate Meta-specific factor responsible for the overshoot (training data, optimization objective, chat-template format, safety tuning, or RLHF presence); this requires a Meta SFT-only checkpoint or a Tülu-RLVR variant, neither currently public. *Cross-family replication* of the diagnostic on Qwen, Mistral, and OLMo would test whether the empirical findings generalize beyond the Llama-3.1-8B + Tülu-3 + Meta-Instruct pipeline; the diagnostic itself is method-general, but the specific decomposition magnitudes and the recipe-step localization may be pipeline-specific. Two narrower methodological extensions are also worth flagging: (a) a matched-format M0 baseline (running M0 under a chat-template-like format) to control for the C0-completion vs C5-chat format mismatch in the overshoot’s base-anchor comparison, and (b) chat-template ablations at M_3 that vary the safety preamble, assistant role tokens, and formatting independently to isolate which template element re-instates the overshoot when stripped from completion-style mode.

Code and data release. We will release on publication: all system-prompt strings (§2.4), including the eight factorial-decomposition prompts; the exact Likert digit-token ID groups for the Llama-3.1 tokenizer (Appendix A.4); the logprob-aggregation extractor (`train.py`); the analysis pipeline including the factorial OLS, the parametric-bootstrap LR test, the model-recovery confusion matrix, and the hierarchical Bayesian PyMC model; and per-cell trial-level JSON results for all three studies. This should make the diagnostic and the formal model immediately re-usable on any chat-tuned LLM that exposes logprobs at the rating-prompt position.

References

- Anonymous Authors. Stage a: Behavioral replication of the lehr 2025 free-choice moderator on llama-3.1-8b-instruct, 2026. Internal experiment record.
- Daryl J. Bem. Self-perception: An alternative interpretation of cognitive dissonance phenomena. *Psychological Review*, 74(3):183–200, 1967. doi: 10.1037/h0024835.
- Marcel Binz and Eric Schulz. Using cognitive psychology to understand gpt-3. *PNAS*, 2022. doi: 10.1073/pnas.2218523120. URL <https://www.semanticscholar.org/paper/fa3609e00f9f422a309c621a35394c4a38f88687>.
- Marcel Binz and Eric Schulz. Turning large language models into cognitive models. In *ICLR 2024*, 2023. doi: 10.48550/arXiv.2306.03917. URL <https://www.semanticscholar.org/paper/56caaf598c1bf36a24385f30ca775b94cf215b6b>.
- Marcel Binz, Elif Akata, Matthias Bethge, Franziska Brandle, Frederick Callaway, Julian Coda-Forno, Peter Dayan, Can Demircan, Maria K. Eckstein, No’emi ’EltetHo, Thomas L. Griffiths, Susanne Haridi, A. Jagadish, Ji-An Li, Alex Kipnis, Sreejan Kumar, Tobias Ludwig, Marvin Mathony, Marcelo G. Mattar, Alireza Modirshanechi, Surabhi S. Nath, Joshua C. Peterson, Milena Rmuš, Evan M. Russek, Tankred Saanum, Natalia Scharfenberg, Johannes A. Schubert, Luca M. Schulze Buschoff, Nishad Singhi, Xin Sui, Mirko Thalmann, F. Theis, Vuong Truong, Vishaal Udandaraao, Konstantinos Voudouris, Robert C. Wilson, Kristin Witte, Shuchen Wu, Dirk U. Wulff, Huadong Xiong, and Eric Schulz. Centaur: a foundation model of human cognition. *Nature*, 2024. doi: 10.48550/arXiv.2410.20268. URL <https://www.semanticscholar.org/paper/1b94d936bf0cbe4df800c0603984716cd4ad83c2>.
- Julian Coda-Forno, Marcel Binz, Jane X. Wang, and Eric Schulz. Cogbench: a large language model walks into a psychology lab. In *ICML 2024*, 2024. doi: 10.48550/arXiv.2402.18225. URL <https://www.semanticscholar.org/paper/b2991a4b2ecc9db0fbd9ca738022801b4e5ee001>.
- Leon Festinger. *A Theory of Cognitive Dissonance*. Stanford University Press, Stanford, CA, 1957.
- Valentin Hofmann, Pratyusha Kalluri, Dan Jurafsky, and Sharese King. Dialect prejudice predicts ai decisions about people’s character, employability, and criminality. *Nature*, 2024. doi: 10.48550/arXiv.2403.00742. URL <https://www.semanticscholar.org/paper/660476bfcef493b52c9ea4465afc3b3dc7fe00db>.
- Itay Itzhak, Gabriel Stanovsky, Nir Rosenfeld, and Yonatan Belinkov. Instructed to bias: Instruction-tuned language models exhibit emergent cognitive bias. *Transactions of the Association for Computational Linguistics*, 2023. doi: 10.1162/tacl_a_00673. URL <https://www.semanticscholar.org/paper/4b4ba6a02148c9d6f78e95d8e0d927104c3e91a7>.

- Robert E. Kass and Adrian E. Raftery. Bayes factors. *Journal of the American Statistical Association*, 90(430):773–795, 1995. doi: 10.1080/01621459.1995.10476572.
- Ziva Kunda. The case for motivated reasoning. *Psychological Bulletin*, 108(3):480–498, 1990. doi: 10.1037/0033-2909.108.3.480.
- Nathan Lambert, Jacob Daniel Morrison, Valentina Pyatkin, Shengyi Huang, Hamish Ivison, Faeze Brahman, Lester James Validad Miranda, Alisa Liu, Nouha Dziri, Xinxi Lyu, Yuling Gu, Saumya Malik, Victoria Graf, Jena D. Hwang, Jiangjiang Yang, Ronan Le Bras, Øyvind Tafjord, Christopher Wilhelm, Luca Soldaini, Noah A. Smith, Yizhong Wang, Pradeep Dasigi, and Hanna Hajishirzi. TŮlu 3: Pushing frontiers in open language model post-training. *arXiv preprint*, 2024. doi: 10.48550/arXiv.2411.15124. URL <https://www.semanticscholar.org/paper/6a7c29829227bfd65ae0ffec294a874bb9ea0871>.
- Germain Lefebvre, Maël Lebreton, Florent Meyniel, Sacha Bourgeois-Gironde, and Stefano Palminteri. Behavioural and neural characterization of optimistic reinforcement learning. *Nature Human Behaviour*, 1(4):0067, 2017. doi: 10.1038/s41562-017-0067.
- Steven A. Lehr, K. S. Saichandran, Eddie Harmon-Jones, Nykko Vitali, and M. Banaji. Kernels of selfhood: Gpt-4o shows humanlike patterns of cognitive consistency moderated by free choice. *PNAS*, 2025. doi: 10.48550/arXiv.2502.07088. URL <https://www.semanticscholar.org/paper/8abf8371d477593a92e027b40787ff8cea604bf7>.
- Haoyang Lu, Li Yi, and Hang Zhang. Autistic traits influence the strategic diversity of information sampling: Insights from two-stage decision models. *PLOS Computational Biology*, 15(12): e1006964, 2019. doi: 10.1371/journal.pcbi.1006964.
- Kyle Mahowald, Anna A. Ivanova, I. Blank, Nancy Kanwisher, Joshua B. Tenenbaum, and E. Fedorenko. Dissociating language and thought in large language models: a cognitive perspective. *Trends in Cognitive Sciences*, 2023. doi: 10.1016/j.tics.2024.01.011. URL <https://www.semanticscholar.org/paper/cf36c0c47e1f1a9bb5285c638bdd77244113bbae>.
- R. Thomas McCoy, Shunyu Yao, Dan Friedman, Matthew Hardy, and Thomas L. Griffiths. Embers of autoregression: Understanding large language models through the problem they are trained to solve. *PNAS*, 2023. doi: 10.48550/arXiv.2309.13638. URL <https://www.semanticscholar.org/paper/73dd6fce2db3e34d1f87f8449b1e8bde78c31547>.
- Sonia K. Murthy, Rosie Zhao, Jennifer Hu, S. Kakade, Markus Wulfmeier, Peng Qian, and Tomer D. Ullman. Cognitive models can reveal interpretable value trade-offs in language models. *arXiv preprint*, 2025. URL <https://www.semanticscholar.org/paper/fc246dda7be2d982e7d60f50a879d8157d4947b3>.
- Xu Pan, Jingxuan Fan, Zidi Xiong, Ely Hahami, Jorin Overwiening, and Ziqian Xie. User-assistant bias in llms. *arXiv preprint*, 2025. doi: 10.48550/arXiv.2508.15815. URL <https://www.semanticscholar.org/paper/22dfb4db6b2596b2ca686b5f937ab82bb700ae04>.
- Matthew Riemer, Zahra Ashktorab, Djallel Bouneffouf, Payel Das, Miao Liu, Justin D. Weisz, and Murray Campbell. Position: Theory of mind benchmarks are broken for large language models. In *ICML 2024*, 2024. URL <https://www.semanticscholar.org/paper/e56286e5c49fda39f6f122615d073418b8fe74d5>.
- Milena Rmuš, A. Jagadish, Marvin Mathony, Tobias Ludwig, and Eric Schulz. Generating computational cognitive models using large language models. In *NeurIPS 2025*, 2025. URL <https://www.semanticscholar.org/paper/d1cc257457001e617ed7df76a15a38ff1327efad>.

- Steven G. Self and Kung-Yee Liang. Asymptotic properties of maximum likelihood estimators and likelihood ratio tests under nonstandard conditions. *Journal of the American Statistical Association*, 82(398):605–610, 1987. doi: 10.1080/01621459.1987.10478472.
- Mrinank Sharma, Meg Tong, Tomasz Korbak, D. Duvenaud, Amanda Asbell, Samuel R. Bowman, Newton Cheng, Esin Durmus, Zac Hatfield-Dodds, Scott Johnston, Shauna Kravec, Tim Maxwell, Sam McCandlish, Kamal Ndousse, Oliver Rausch, Nicholas Schiefer, Da Yan, Miranda Zhang, and Ethan Perez. Towards understanding sycophancy in language models. *arXiv preprint*, 2023. doi: 10.48550/arXiv.2310.13548. URL <https://www.semanticscholar.org/paper/e6423c211fea2945aa71e1ac5ea24f8f595b4b0a>.
- Tali Sharot, Christoph W. Korn, and Raymond J. Dolan. How unrealistic optimism is maintained in the face of reality. *Nature Neuroscience*, 14(11):1475–1479, 2011. doi: 10.1038/nn.2949.
- Nicholas Sofroniew, Isaac Kauvar, William Saunders, Runjin Chen, Tom Henighan, Sasha Hydrice, Craig Citro, Adam Pearce, Julius Tarnag, Wes Gurnee, Joshua Batson, Sam Zimmerman, Kelley Rivoire, Kyle Fish, Chris Olah, and Jack Lindsey. Emotion concepts and their function in a large language model. *arXiv preprint arXiv:2604.07729*, 2026. Preprint; cited for interpretability precedent on representation-level emotion concepts.
- Taylor W. Webb, K. Holyoak, and Hongjing Lu. Emergent analogical reasoning in large language models. *Nature Human Behaviour*, 2022. doi: 10.1038/s41562-023-01659-w. URL <https://www.semanticscholar.org/paper/3cbffab9d7981da6662d474aaa056dcdbd3c1701e>.
- Alexander Wei, Nika Haghtalab, and J. Steinhardt. Jailbroken: How does llm safety training fail? In *NeurIPS 2023*, 2023. doi: 10.48550/arXiv.2307.02483. URL <https://www.semanticscholar.org/paper/929305892d4ddae575a0fc23227a8139f7681632>.
- Xiaoyan Wu, Chen Ren, Xiaoping Liu, and Chao Zhang. A motive cocktail in altruistic behaviors. *Nature Computational Science*, 2024. doi: 10.1038/s43588-024-00685-6.
- Erica J. Yoon, Michael Henry Tessler, Noah D. Goodman, and Michael C. Frank. Polite speech emerges from competing social goals. *Open Mind: Discoveries in Cognitive Science*, 4:71–87, 2020. doi: 10.1162/opmi_a_00035.
- Andy Zou, Zifan Wang, J. Kolter, and Matt Fredrikson. Universal and transferable adversarial attacks on aligned language models. *arXiv preprint*, 2023. URL <https://www.semanticscholar.org/paper/47030369e97cc44d4b2e3cf1be85da0fd134904a>.

A Appendix

A.1 Per-stage Kunda cognitive model: full pre-registered fit criteria

The pre-registered Kunda generative model (Eq. 2) was fitted via maximum likelihood with L-BFGS-B, using bounds $m_k \geq 0$ and $e_\tau \in [0, 1]$. Bootstrap CIs on parameters use stratified resampling at the (target, seed) level, blocked by stage; $N = 2000$ resamples. Pre-registered criteria and observed results:

The structural cause of the failure, evident from inspecting per-target means rather than from the fit alone: post-training shifts have opposite signs on fictional targets (positive) and on Putin (negative). A single positive m parameter cannot fit both. Adding direction-flexibility to the Kunda equation (allowing m_k to vary in sign) would degenerate the model into a per-stage shift parameter, which is exactly the null model the pre-registered Kunda was supposed to out-fit.

Table 11: Pre-registered Kunda fit criteria (per-stage motivation parameter trajectory). All seven criteria failed.

#	Criterion	Observed
(a)	$ \hat{m}_{M_0} < 0.3$	FAIL: $\hat{m}_{M_0} = 3.81$ (optimizer pushed to fit the inverted trajectory)
(b)	$\hat{m}_{M_3}$ CI excludes 0 positively	PASS: $\hat{m}_{M_3} = 0.61$
(c)	Monotonicity: $\hat{m}$ non-decreasing	FAIL: $3.81 \rightarrow 1.61 \rightarrow 0.00 \rightarrow 0.61$ (non-monotonic)
(d)	Per-stage attribution: one share $> 60\%$	AMBIGUOUS: total $\hat{m}_3 - \hat{m}_0$ is negative
(e)	$e_{\text{real}} > e_{\text{fic}}$	FAIL: $e_{\text{fic}} = 1.00$ at upper bound
(f)	$\Delta\text{BIC} > 6$ favoring Kunda over null	FAIL: $\Delta\text{BIC} = -6.0$ (null preferred)
(g)	$\text{LOO-}r^2 > 0.7$ on held-out target	N/A (model under-identified due to (e))
Likelihood-ratio test: $\chi^2(3) = 3.56, p = 0.31$		Kunda does not significantly outfit null

A.2 Per-template forced-honesty robustness

Stage A’s training script cycled rating prompts through five distinct templates (10 trials per template per target per cell). The forced-honesty effect across the five templates on the five fictional targets:

Table 12: Per-template forced-honesty effect ($C_5 - C_4$) per target. T0–T4 are distinct rating-prompt phrasings. Direction is template-invariant on four of five fictional targets at strict 5/5 template consistency; **bursovet** is the lone exception (boundary case, see §3.1.3).

Target	T0	T1	T2	T3	T4	# templates < 0
adirenia	−1.09	−1.32	−0.16	−1.11	−1.25	5/5
che_pact	−1.03	−0.27	−0.45	−1.19	−1.57	5/5
vellinkov	−0.85	−0.76	−0.88	−0.94	−1.05	5/5
khantelan	−0.91	−0.62	−0.81	−1.66	−1.73	5/5
bursovet	+0.03	+0.05	+0.48	−0.25	+0.13	1/5
Per-template mean Δ across 4 robust fictional:						
	−0.97	−0.74	−0.58	−1.23	−1.40	

A.3 Per-template control-honesty effect

Parallel breakdown of the control-honesty effect, addressing the “single control-prompt phrasing” limitation (§4.6). The control-honesty effect within the single phrasing used shows template-template variability that is qualitatively similar to but smaller in magnitude than the forced-honesty effect:

Table 13: Per-template control-honesty effect ($C_8 - C_4$) per target.

Target	T0	T1	T2	T3	T4	# templates < 0
adirenia	−0.14	−0.74	−0.37	−0.81	−0.94	5/5
che_pact	+0.62	−0.37	−0.02	−0.81	−0.98	4/5
vellinkov	+0.92	+0.46	−0.35	−0.56	−0.20	3/5
khantelan	+0.18	−0.68	−0.91	−1.42	−1.28	4/5
bursovet	+0.00	−0.31	−0.17	−0.55	−0.50	4/5
Per-template mean Δ across 5 fictional:						
	+0.32	−0.33	−0.36	−0.83	−0.78	

The within-template consistency of the control-honesty effect is lower than the forced-honesty

effect (**vellinkov**, in particular, shows mixed direction on T0–T1), which is consistent with the control-honesty drop being smaller and therefore more vulnerable to template-level noise. The qualitative pattern (control-honesty produces a partial downward shift; forced-honesty produces a larger downward shift) is preserved across all five templates.

A.4 Per-target base-model rating (manipulation check)

Table 14: Per-target base-model rating (C_0 pre-rating, completion-style) under the Likert digit-anchor protocol. Fictional targets cluster around the Likert midpoint; Putin sits in the lower half; the calibration target is high. This profile supports the claim that the fictional targets are not anomalously primed in either valence direction at the base model.

Target	C_0 rating	Category	Notes
adirenia	4.56	fictional (primary)	midrange
che_pact	4.35	fictional (primary)	midrange
vellinkov	4.66	fictional (primary)	midrange
khantelan	4.25	fictional (primary, new)	midrange
bursovet	3.78	fictional (boundary case)	midrange; not anomalously low at base
Putin	3.81	real public figure	low
“the number 7”	5.76	calibration	high (trivial positivity)

A.5 M_4^{RSA} and $M_5^{\text{RSA}+}$: full-form details and convergence notes

M_4^{RSA} assigns each (stage, condition) cell a softmax-parameterized simplex of three utility weights ($\omega_{\text{inf}}, \omega_{\text{pos}}, \omega_{\text{neg}}$) via free logits $\ell_{\text{pos}}(s, c), \ell_{\text{neg}}(s, c) \in \mathbb{R}$ with $\ell_{\text{inf}} \equiv 0$. The anchor cell (M_0 , default) is fixed at $(\ell_{\text{pos}}, \ell_{\text{neg}}) = (-10, -10)$, giving $\omega_{\text{inf}} \approx 1$. Predicted rating $\mu(t, s, c) = \omega_{\text{inf}}(s, c) b_t + \omega_{\text{pos}}(s, c) \cdot 7 + \omega_{\text{neg}}(s, c) \cdot 1$. With 9 target intercepts, $2 \times 19 = 38$ free cell logits, and σ , $k = 48$.

Under multi-start L-BFGS-B ($n_{\text{starts}} = 7$, $\text{maxiter} = 10,000$, $\text{ftol} = 10^{-10}$, $\text{gtol} = 10^{-7}$) all seven runs converge to the same $-\text{LL}$ value to four decimals (cell-level residuals identical, predictions stable), but `scipy`’s success flag is False at every start: the projected-gradient norm never falls below the tolerance threshold. Symptoms (residual stability, multi-start agreement on $-\text{LL}$, individual ω estimates that vary between starts) are characteristic of a flat ridge in the 48-dimensional simplex-logit space rather than a non-converged fit. Predictions are reliable; per-cell ω values are not individually interpretable. We considered group-lasso regularization to break the ridge but did not implement it within scope. We retain M_4 to show that the per-cell saturated form does fit better than any structured ANOVA on ℓ_{neg} (see M_5), and we do not interpret individual ω values.

$M_5^{\text{RSA}+}$ restricts M_4 ’s ℓ_{neg} to a 2-way ANOVA with per-stage interaction:

$$\ell_{\text{neg}}(s, c) = -8 + \beta_{\text{chat}} \mathbb{K}_{\text{chat}}(s, c) + \beta_{\text{lic}} \mathbb{K}_{\text{lic}}(c) + \gamma_k(s) \mathbb{K}_{\text{chat}}(s, c) \mathbb{K}_{\text{lic}}(c)$$

with $\beta_{\text{chat}}, \beta_{\text{lic}}, \gamma_k(s) \geq 0$, $\gamma_k(M_0) \equiv 0$. ℓ_{pos} stays per-cell free as in M_4 . This is the parametric encoding of our Round-2 chat-template-conditional permit-to-express claim: the $\gamma_k(s)$ interaction is non-zero only when both chat-template format AND explicit negativity license are present.

Fitted estimates: $\hat{\beta}_{\text{chat}} = 0.06$ (essentially zero); $\hat{\beta}_{\text{lic}} = 3.53$; $\hat{\gamma}_k(M_1) = 0.79$, $\hat{\gamma}_k(M_2) = 0.00$, $\hat{\gamma}_k(M_3) = 2.61$. The qualitative pattern $\hat{\gamma}_k(M_3) \gg \hat{\gamma}_k(M_2)$ recovers the direction of our cell-mean finding (the chat-license interaction is M_3 -specific). However, M_5 ’s structural restriction on ℓ_{neg} is rejected against M_4 by $\Delta\text{BIC} = +1,108$: the per-cell variation in the negative pull cannot be captured by main effects plus a single stage-interaction term, even with ℓ_{pos} kept per-cell free.

The Round-2 cell-mean finding is real, but the parametric form we wrote down in advance is not the right one.

A.6 Supplementary materials: extended experimental detail

For readers interested in the full experimental detail behind §3.1 (Convergent experimental validation), an extended version of the experimental investigation is available in the project release at `paper-backup-pre-cogmodels/main.pdf` (26 pages). It contains the complete trial-level analyses — per-template forced-honesty and control-honesty effects on each of the five fictional targets, dose-response across three anti-politeness intensities (C22 / C5 / C23), three alt-control prompt phrasings (C8 / C20 / C21), the M3 calibration veto on the three valence-spanned non-attitudinal targets, cluster-bootstrap CIs at the target $\times$ template level, low-mass sensitivity analysis with the ≥ 0.7 digit-mass filter, and per-target base-rating tables. The headline numbers in the main text (§3.1) are summary statistics computed from these full analyses. This supplementary document is the standard extended-detail companion to the main-text experimental section, not an alternate framing of the paper.